A computationally efficient SPH framework for unsaturated soils and its application to predicting the entire rainfall-induced slope failure process

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ABSTRACT

We present the first and fully validated SPH model to tackle coupled flow-deformation problems in unsaturated porous media that undergo large deformation and post-failure behaviour. Unlike the commonly adopted double-layer SPH framework for saturated soils, this paper presents a three-phase single-layer SPH model capable of predicting anisotropic seepage flows through porous media and their complete time-dependent transition from unsaturated to saturated states, as well as their influence on the mechanical behaviour of the porous media and vice-versa. The mathematical framework is developed based on Biot's mixture theory and discretised using our recently developed novel SPH approximation scheme for the second derivatives of a field quantity. The soil is modelled using a suction-dependent elastoplastic constitutive model, expressed in terms of effective stress and suction. In addition, an adaptive two-timescale scheme is proposed for the first time to address existing challenges in solving coupled-flow large deformation problems that involve a significant difference in the timescale required for the solid and fluid phases. The capability of the proposed SPH model was demonstrated through fundamental consolidation tests and a large-scale rainfall-induced slope failure experiment. Very good agreements with theoretical solutions and experimental results are achieved, suggesting that the proposed SPH model can be readily extended to solve a wide range of large-scale geotechnical applications involving coupled unsaturated seepage-deformation problems.

Keywords: SPH, rainfall-induced slope failure, large deformation, coupled flow-deformation analysis, Unsaturated media.

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INTRODUCTION

The design of many geotechnical applications requires the knowledge of unsaturated soils, and in many cases, such applications involve large deformations and failures of these materials. Rainfall-induced slope failure is one of these applications that has been extensively studied in the past (Alonso, 2021; Borja & White, 2010; Switała & Wu, 2018; Yerro *et al.*, 2015). When rainfall water infiltrates into the ground, the accumulated rainwater can cause the groundwater level to rise, leading to a reduction in soil suctions and shear strength of soils, threatening the performance and safety of geo-infrastructures, or in the worst-case scenarios, causing severe damages and failures to these infrastructures. To predict these problems, the conventional Finite Element Method (FEM) is often a popular choice owing to its capability to provide robust and accurate solutions for small deformation analyses. However, when it comes to large deformation and failure predictions of geomaterials, FEM is well-known for suffering from mesh distortion issues (Bui & Nguyen, 2021). Recent advances in FEM modelling led to several FEM-based modelling approaches that could address issues associated with mesh distortion, including the Particle Finite Element Method (Jin *et al.*, 2020; Oñate *et al.*, 2004) or the Material Point Method (Alonso, 2021; Bandara & Soga, 2015; Yerro *et al.*, 2015), to name a few. Nevertheless, there is still a need to develop a more robust computational approach that can predict fully-coupled seepage-induced large deformation and failure behaviour of geomaterials.

Smoothed Particle Hydrodynamics (SPH) is one of the oldest truly mesh-free methods, which was originally developed for astrophysical applications (Gingold & Monaghan, 1977; Lucy, 1977). Since its first successful application for the modelling of the elastoplastic behaviour of geomaterials (Bui, Fukagawa, *et al.*, 2008), SPH has been widely used to solve a range of challenging geotechnical problems, for example, granular flows (Bui *et al.*, 2008;Chalk *et al.*, 2020; Nguyen *et al.*, 2017; Peng *et al.*, 2015, 2016; Yang *et al.*, 2020; Fávero Neto & Borja, 2018), slope failures (Blanc & Pastor, 2012; Bui *et al.*, 2011a; Bui & Fukagawa, 2013; He *et al.*, 2018; Pastor *et al.*, 2014; Zhao *et al.*, 2019), earthquake-induced soil slope failure (Bui *et al.*, 2010; Bao *et al.*, 2020; Chen & Qiu, 2014), soil-structure interactions (Bui *et al.*, 2014; Bui *et al.*, 2008; Sheikh *et al.*, 2020; Yang *et al.*, 2021; Zhan *et al.*, 2020), desiccation cracking in soils (Tran *et al.*, 2019, 2020) and rock fractures (del Castillo *et al.*, 2021; Gharehdash *et al.*, 2020; Wang *et al.*, 2019, 2020). Besides, for coupled fluid-solid interaction problems in geomechanics, there exist two SPH approaches to solve governing equations for fully saturated soils: multi-layer particle and single-layer particle approaches (Bui & Nguyen, 2021). In the first SPH approach, a saturated porous medium is represented by two layers of Lagrangian particles, over which the governing equations of each phase are solved separately (Bui *et al.*, 2007; Bui & Nguyen, 2017; Pastor *et al.*, 2018). In the second SPH approach, the saturated porous medium is represented by a single layer of Lagrangian particles, each of which carries the information of both water and solid phases, and the fully coupled governing equations of this two-phase system are solved on a single set of SPH particles (Bui & R. Fukagawa, 2009; Pastor *et al.*, 2009). Both multi-layer and single-layer approaches have their own advantages in specific applications. For example, the multi-layer method is more suitable for problems where explicit interactions between the solid and fluid phases are required (e.g. overtopping or internal piping failure behaviour of saturated soils). On the other hand, the single-layer approach is considered to be computationally cheaper and thus is more suitable for solving large-scale boundary values applications. Recently, based on the single-layer SPH particle approach, a robust SPH computational framework for solving transient seepage flow through anisotropic unsaturated porous media has been fully developed (Lian *et al.*, 2021). However, its application to solve fully coupled fluid-solid interaction problems in unsaturated soils is not attested, and to the best of the authors' knowledge, no SPH attempts to solve these fully coupled problems in unsaturated soils are reported in the literature until today. To address this issue, this paper aims to further advance the application of the single-layer SPH approach to solving complex coupled flow-deformation problems in unsaturated porous media.

GOVERNING EQUATIONS

Mass conservation equations

The mass balance equation for each phase in a unit volume of an unsaturated soil mixture consisting of air, water and solid phases can be written as follows (Bui & Nguyen, 2017):

$$\frac{d^\alpha \bar{\rho}_\alpha}{dt} + \bar{\rho}_\alpha \nabla \cdot \mathbf{v}_\alpha = 0 \tag{1}$$

where α denotes phase component, with s, l and a being solid, water and air phases, respectively; $\bar{\rho}_\alpha = n_\alpha \rho_\alpha$ is the partial densities for each component with ρ_α being the intrinsic mass density of each constituent; and n_α is the volume fraction of each phase and related to porosity (n) and degree of saturation (S_r) as follows (see Fig. 1): $n_s = 1 - n$, $n_l = nS_r$ and $n_a = n(1 - S_r)$; $\mathbf{v}_\alpha$ is the velocity vectors of phase α in the mixture; and d^α/dt denotes the material derivative of a field quantity on phase α.

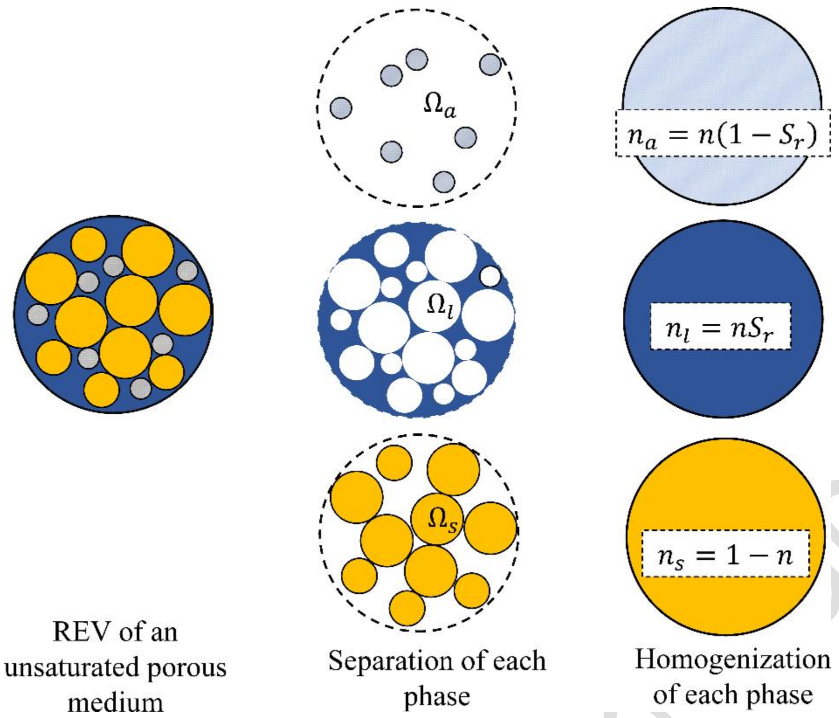

Fig. 1. Schematic diagram of a representative element volume (REV) of an unsaturated porous medium and its homogenisation strategies for continuum modelling.

Expanding Eq. (1) for each phase component and rewriting all resulting equations on the same solid phase reference framework, the following equations can be obtained after enforcing the incompressibility condition for solid grains (Lian *et al.*, 2021):

$$\frac{d^s n}{dt} = (1-n)\nabla \cdot \mathbf{v}_s \tag{2}$$

$$\frac{n_l}{\rho_l}\left(\frac{d^s \rho_l}{dt} + \overline{\mathbf{w}}_{ls} \cdot \nabla \rho_l\right) + n\frac{d^s S_r}{dt} + S_r \nabla \cdot \mathbf{v}_s + \nabla \cdot (nS_r \overline{\mathbf{w}}_{ls}) = 0 \tag{3}$$

$$\frac{n_a}{\rho_a}\left(\frac{d^s \rho_a}{dt} + \overline{\mathbf{w}}_{as} \cdot \nabla \rho_a\right) - n\frac{d^s S_r}{dt} + (1 - S_r)\nabla \cdot \mathbf{v}_s + \nabla \cdot \left(n_a \overline{\mathbf{w}}_{as}\right) = 0 \tag{4}$$

where and $\overline{\mathbf{w}}_{\alpha\beta} = \mathbf{v}_\alpha - \mathbf{v}_\beta$ is the relative velocity between two phases.

By enforcing the linear relationship between (p_l) and volumetric strain of water ($\varepsilon_l^{\mathrm{v}}$) and adopting an ideal gas law for the rate of change of fluid and gas density, the following evolution equations for the pore-water and pore-air pressures can be obtained (Lian *et al.*, 2021):

$$n_l \hat{\beta}\frac{d^s p_l}{dt} + \frac{1}{\rho_l} n_l \overline{\mathbf{w}}_{ls} \cdot \nabla \rho_l + n\frac{d^s S_r}{dt} + S_r \nabla \cdot \mathbf{v}_s + \nabla \cdot (n_l \overline{\mathbf{w}}_{ls}) = 0 \tag{5}$$

$$\frac{n_a}{\rho_a}\left(\frac{1}{RT}\frac{d^s p_a}{dt} + \overline{\mathbf{w}}_{as} \cdot \nabla \rho_a\right) - n\frac{d^s S_r}{dt} + \frac{n_a}{n}\nabla \cdot \mathbf{v}_s + \nabla \cdot (n_a \overline{\mathbf{w}}_{as}) = 0 \tag{6}$$

where $\hat{\beta} = 1/K_l^{sat}$ is the compressibility of the pore fluid with K_l^{sat} being the bulk modulus; R is the specific gas constant for the dry air; T is the temperature. To close the above equations, the definition of relative velocities between water and solid phases ($n_l \overline{\boldsymbol{w}}_{ls}$) and between air and solid phases ($n_a \overline{\boldsymbol{w}}_{as}$) is required. These can be achieved by considering the momentum balances of the unsaturated soil mixture, which are derived in the next section.

Momentum balance equations

The behaviour of unsaturated porous media is governed by the interaction between solid skeleton and pore-fluids, each of which is considered as a homogenised continuum that follows its own governing equations. Thus, the general form of the linear momentum balance equation for each phase component in the three-phase mixture can be written as follows:

$$\bar{\rho}_\alpha \frac{d^\alpha \mathbf{v}_\alpha}{dt} = \nabla \cdot \bar{\boldsymbol{\sigma}}_\alpha + \bar{\rho}_\alpha \mathbf{b} - \sum \boldsymbol{R}^{\alpha\beta} \tag{7}$$

where $\bar{\boldsymbol{\sigma}}_\alpha = n_\alpha \boldsymbol{\sigma}_\alpha$ is the partial stresses of each phase in the mixture; $\mathbf{b} = [0 \quad -g]$ is the body force vector with g being the gravitational acceleration; and $\mathbf{R}^{\alpha\beta}$ is the drag force vector between phases α and β and takes the following form:

$$\mathbf{R}^{\alpha\beta} = \frac{\bar{\rho}_\alpha g}{\mathbf{k}_\alpha} n_\alpha \overline{\mathbf{w}}_{\alpha\beta} - p_\alpha \nabla n_\alpha \tag{8}$$

where $\mathbf{k}_\alpha$ is the permeability coefficient matrix of phase α.

By expanding Eq. (7) and neglecting the interaction force between water and air phases, the relative velocities between the pore fluids (i.e. water and air) and the solid skeleton can be written as follows, respectively (Lian *et al*., 2021):

$$\overline{\mathbf{w}}_{ls} = \frac{\mathbf{k}_l}{n_l \gamma_l}\left(-\nabla p_l + \rho_l \mathbf{b} - \rho_l \frac{d^s \mathbf{v}_s}{dt}\right) \tag{9}$$

$$\overline{\mathbf{w}}_{as} = \frac{\mathbf{k}_a}{n_a \gamma_a}\left(-\nabla p_a + \rho_a \mathbf{b} - \rho_a \frac{d^s \mathbf{v}_s}{dt}\right) \tag{10}$$

On the other hand, the momentum equation for the entire unsaturated soil mixture can be derived by summing the momentum equation for each phase and neglecting the relative accelerations among the solid, water and air phases, yielding:

$$\frac{d^s \mathbf{v}_s}{dt} = \frac{1}{\rho_t} \nabla \cdot \boldsymbol{\sigma} + \mathbf{b} \tag{11}$$

where $\boldsymbol{\sigma} = \boldsymbol{\sigma}' - [S_r p_l + (1 - S_r) p_a]\mathbf{I}$ is the total stress tensor with $\boldsymbol{\sigma}'$ being the effective stress; and ρ_t is the total density of the mixture. The assumption of neglecting the relative acceleration is expected to produce a minimal influence on the numerical results because the fluid acceleration is usually much smaller than that of the solid phase (Zienkiewicz *et al.* 1999). Furthermore, this assumption is valid for most practical engineering applications because the permeability of unsaturated porous media is generally very low. It is also worth noting here that, different from the double-layer SPH approach (Bui & Nguyen, 2017), which separately solves the momentum equation for each phase to determine the motion of each layer of particles, the present single-layer SPH model solves the motion of the solid phase using the total stress of the entire mixture. Therefore, to close the above equation, one needs to define a constitutive model for the effective stress, and this will be detailed in the subsequent section.

Seepage flow equations in deformable unsaturated porous media

The general equations governing the unsaturated seepage flow through deformable porous media can be obtained by combining the mass and momentum conservation equations, and this can be achieved by substituting Eqs. (9)-(10) into Eqs. (5)-(6):

$$n_l\hat{\beta}\frac{d^s p_l}{dt} + \frac{1}{\rho_l}n_l\overline{\mathbf{w}}_{ls}\cdot\nabla\rho_l + n\frac{d^s S_r}{dt} + S_r\nabla\cdot\mathbf{v}_s + \frac{1}{\gamma_l}\nabla \cdot\left[\mathbf{k}_l\left(-\nabla p_l + \rho_l\mathbf{b} - \rho_l\frac{d^s\mathbf{v}_s}{dt}\right)\right] = 0 \tag{12}$$

$$\frac{n_a}{\rho_a}\left(\frac{1}{RT}\frac{d^s p_a}{dt} + \overline{\mathbf{w}}_{as}\cdot\nabla\rho_a\right) - n\frac{d^s S_r}{dt} + \frac{n_a}{n}\nabla\cdot\mathbf{v}_s + \frac{1}{\gamma_a}\nabla \cdot\left[\mathbf{k}_a\left(-\nabla p_a + \rho_a\mathbf{b} - \rho_a\frac{d^s\mathbf{v}_s}{dt}\right)\right] = 0 \tag{13}$$

Equations (12) and (13) describe the time evolutions of pore-water and pore-air pressures in the unsaturated porous media, respectively, and in principle, can be solved using any appropriate numerical method once an appropriate hydraulic constitutive model is provided. Together with Eq. (11), they form the $u - p_a - p_l$ formulations for the coupled flow-deformation analysis in unsaturated soils. These governing equations can be numerically solved using appropriate time integration schemes to obtain the evolution of air and water pressures as well as the solid motion in the unsaturated soil mixture. Nevertheless, for many practical engineering applications, the pore-air pressure usually remains constant or can be treated as a prescribed function (Sheng *et al.*, 2003). Therefore, in this paper, in an attempt to establish the first SPH framework to describe the coupled flow-deformation in unsaturated porous media, the evolution of pore-air pressure is neglected (i.e. $p_a = 0$). Furthermore, we also assumed that the spatial gradient of fluid density is negligible, given the fluid phase is incompressible, although its compressibility is controlled by the water bulk modulus in the present numerical framework. Again, it is expected that this assumption produces a negligible influence on the numerical results. Therefore, the above general equations for the transient seepage flow through deformable unsaturated porous media reduces to:

$$\frac{d^s h}{dt} = \frac{1}{\tilde{C}_{Sr}} \left\{ \nabla \cdot [\mathbf{k} \nabla (h + z)] - S_r \nabla \cdot \mathbf{v}_s + \frac{1}{g} \nabla \cdot \left(\mathbf{k}_l \, \frac{d^s \mathbf{v}_s}{dt} \right) \right\} \tag{14}$$

where $\tilde{C}_{Sr} = (C_l + C_s)$ are the summation of the specific storage term $C_l(K_l^{sat}) = \gamma_l / n_l \hat{\beta}$ and specific moisture term $C_s(h, S_r)$, which can be determined from a hydraulic constitutive model or a soil-water characteristic curve (SWRC); z is the elevation; and $h = p_l/\gamma_l$ is the water pressure head. The above seepage flow equation shows strong couplings between the fluid flow and solid deformation, which play a critical role in modelling coupled problems in unsaturated porous media such as rainfall-induced landslides. To close this seepage flow equation, a hydraulic constitutive model is required and will be detailed in the next section. It is noted that the superscript ^{s} will be dropped in the rest of this paper to simplify the mathematical expressions.

Hydraulic constitutive model

Any existing hydraulic constitutive model or SWRC can be incorporated into the present framework. Among many existing SWRC models (Fredlund & Xing, 1994; van Genuchten, 1980; Kosugi, 1996), the van Genuchten model is perhaps the most popular one, thus is adopted in this paper. The van Genuchten SWRC and its corresponding permeability function are given as:

$$S_r = S_{res} + (S_{sat} - S_{res})[1 + (g_a|-h|)^{g_n}]^{g_c} \tag{15}$$

$$\mathbf{k} = \boldsymbol{k}_{sat}(e) {S_e}^{g_l} \left[1 - \left(1 - {S_e}^{-\frac{1}{g_c}} \right)^{-g_c} \right]^2 \tag{16}$$

where S_{res} is the residual degree of saturation; S_{sat} is the maximum degree of saturation; S_e is the effective degree of saturation defined as $S_e = (S_r - S_{res})/(S_{sat} - S_{res})$; g_a, g_n and g_c are three empirical parameters with $g_c = (1 - g_n)/g_n$; and $\boldsymbol{k}_{sat}(e)$ is the saturated water permeability tensor, which depends on the void ratio (Oka *et al.*, 2019):

$$\boldsymbol{k}_{sat}(e) = \boldsymbol{k}_{sat}^0 \exp\left(\frac{e - e_0}{C_k} \right) \tag{17}$$

where $\boldsymbol{k}_{sat}^0 = k^{sat} \delta^{mn}$ with m and n indicating coordinate directions; e and e_0 are the current and initial void ratios, respectively; and C_k is the Kozeny-Carman coefficient. Finally, to close the fully coupled flow-deformation governing equations for unsaturated porous media, a suction-dependent constitutive model is required to link the variation of soil suctions to shear strength of soil, and this will be given in the next section.

A suction dependent elastic-plastic constitutive model

Any existing advanced elastic-plastic constitutive model for unsaturated soils (Alonso, 1991, 2021; Gens & Alonso, 1992; Khalili & Loret, 2001; Sheng *et al.*, 2003; Wheeler & Sivakumar, 1995) can be incorporated into the above numerical framework to describe the flow-stress-strain relations. In this paper, in an attempt to establish the first single-layer SPH framework for solving coupled flow-deformation of unsaturated soils involving large deformation and failure behaviour, a simple suction-dependent elastic-plastic Drucker-Prager (DP) softening model is considered. The effective stress of this model is defined as follows:

$$\boldsymbol{\sigma}' = \boldsymbol{\sigma} + S_r p_l \mathbf{I}. \tag{18}$$

where **I** is the unit tensor, and the contribution of the pore-air to the effective stress has been neglected. The yield function defined in terms of the effective stress is written as follows:

$$f = \xi_\phi I_1 + \sqrt{J_2} - \kappa_c \tag{19}$$

where I_1 and J_2 are the first and second invariants of the effective stress tensor; ξ_ϕ and κ_c are DP constitutive parameters related to the friction (ϕ) and cohesion (c) of the soil. Under the plane strain condition, these parameters are defined by:

$$\xi_\phi = \frac{tan\phi}{\sqrt{9+12tan\phi^2}} \qquad \kappa_c = \frac{3c}{\sqrt{9+12tan\phi^2}} \tag{20}$$

To describe the evolution of the yield surface with respect to the soil suction, the strength parameters can be redefined as follows:

$$\phi = \phi' + \phi_s \tag{21}$$

$$c = c' + c_s \tag{22}$$

where ϕ' and c' are the effective friction angle and cohesion, respectively; while ϕ_s and c_s represent additional friction and cohesion strengths gained due to the effect of suction when soils are under unsaturated conditions.

In the absence of suction strength, large deformation of soils can cause localised failure and strength softening process. To capture this behaviour, the following strength softening laws for effective friction and cohesion are adopted (Zabala & Alonso, 2011):

$$\phi' = \phi_r + (\phi_p - \phi_r)\exp(-\eta \varepsilon_p^{eq}) \tag{23}$$

$$c' = c_r + (c_p - c_r)\exp(-\eta \varepsilon_p^{eq}) \tag{24}$$

where the subscripts p and r denote the peak and residual effective strengths of the soil, respectively; η is the softening coefficient controlling the rate of shear strength degradation with the accumulated equivalent plastic strain (ε_p^{eq}). On the other hand, under unsaturated conditions, the suction-dependent friction ϕ_s and cohesion c_s can be written as follows (Yerro *et al.*, 2015) :

$$\phi_s = A'\left(\frac{p_s}{p_{atm}}\right) \tag{25}$$

$$c_s = c_{max}\left[1 - \exp\left\{-B'\left(\frac{p_s}{p_{atm}}\right)\right\}\right] \tag{26}$$

where A' and B' are coefficients controlling the rate of variation of friction and cohesion with suction; c_{max} is the maximum cohesion caused by suction; p_{atm} and p_s are the atmospheric pressure and matrix suction, respectively.

The effective stress-strain relationship can be now written as follows:

$$d\boldsymbol{\sigma}' = \mathbf{D}^e : (d\boldsymbol{\varepsilon} - d\boldsymbol{\varepsilon}^p) - \dot{\boldsymbol{\omega}} \cdot \boldsymbol{\sigma}' + \boldsymbol{\sigma}' \cdot \dot{\boldsymbol{\omega}} \tag{27}$$

where $\mathbf{D}^e$ is the elastic stiffness tensor; $d\boldsymbol{\varepsilon}$ is the total strain increment tensor and $\dot{\boldsymbol{\omega}}$ is the spin rate tensor. It is noted in the above stress-strain relation that the Jaumann stress rate has been adopted to maintain the objectivity of the constitutive relation. The selection of the Jaumann stress rate, among several others (e.g. the Truesdell stress rate and the Green–Naghdi stress rate) in the literature, is mainly for ease of numerical implementation. Nevertheless, it is important to note that the Jaumann stress rate should only be used for simulations involving a small time increment or a small deformation/rotation within a given time interval. This condition is usually satisfied in SPH because of the restriction of the CFL condition required to integrate the dynamic governing equations (Bui & Nguyen, 2021), and the position of particles is updated after every time step. A more rigorous treatment to deal with large deformation problems can be achieved by adopting the finite elastoplasticity theory (Song & Borja, 2014; Fávero Neto *et al.*, 2020; Chen & Mizuno, 1990), and the implementation of which is out of the scope of this study.

The plastic strain increment tensor ($d\boldsymbol{\varepsilon}^p$) can be defined by:

$$\boldsymbol{d\varepsilon}^p = d\lambda \frac{\partial g}{\partial \boldsymbol{\sigma}'} \tag{28}$$

where $d\lambda$ is the plastic multiplier; and g is the plastic potential function, which takes the following form:

$$g = \xi_\psi I_1 + \sqrt{J_2} \tag{29}$$

where ξ_ψ takes the same form as that of ξ_ϕ with the friction angle ϕ being replaced by the dilatancy angle ψ. In addition, to properly describe the shearing behaviour of soils at the ultimate shearing stage or critical state (i.e. no volumetric change), the dilatancy angle ψ can be formulated as follows (Nguyen *et al.*, 2020):

$$\psi = \psi_0 \exp\left(-s_f \varepsilon_p^{eq}\right) \tag{30}$$

where ψ_0 is the initial dilation angle; and s_f is a constant controlling the rate of dilatancy reduction. The above constitutive model can be integrated using the semi-implicit time integration scheme (Bui & Nguyen, 2021). Readers are referred to ***Appendix A*** for the detailed derivation of some essential equations required for the stress update of this model.

SPH DISCRETISATIONS OF GOVERNING EQUATIONS

Key SPH formulations

In SPH, a continuum computation domain is discretised into a finite number of particles, each of which carries field variables such as mass, density, velocity and pressure (Bui, Fukagawa, *et al.*, 2008). These variables and their spatial gradients at a given particle are calculated through an interpolation process using information from its surrounding particles and a weighted kernel function. Consider an arbitrary field function $f(\mathbf{x})$ at any point $\mathbf{x}$ in the computational domain, the SPH approximation of this function can be written as follows:

$$\langle f(\mathbf{x}_i)\rangle = \sum_{j=1}^{N} V_j f\left(\mathbf{x}_j\right) W\left(\mathbf{x}_i - \mathbf{x}_j, h_{sml}\right) \tag{31}$$

where the angle bracket $\langle\ \ \rangle$ indicates the kernel approximation operator; V_j is the volume occupied by particle j; $W\left(\mathbf{x}_i - \mathbf{x}_j, h_{sml}\right)$ is a symmetric smoothed (or kernel) function; and h_{sml} is a characteristic length, defining an effective domain Ω of the smoothing function W. Among several popular kernel functions reported in the literature (Bui & Nguyen, 2021; Monaghan, 1985), the Wendland C^2 kernel function is adopted in this paper, which takes the following form:

$$W(q, h_{sml}) = \alpha_d \begin{cases} (1-0.5q)^4(2q+1) & 0 \le q \le 2 \\ 0 & q > 2 \end{cases} \tag{32}$$

where α_d is the dimensional normalising factor defined by $\alpha_d = [5/8h,\ 7/4\pi h^2,\ 21/2\pi h^2]$ for one, two and three-dimensions, respectively; q is the normalised distance defined as $q = \left|\mathbf{x}_i - \mathbf{x}_j\right|/h_{sml}$; and $h_{sml} = 1.3d_0$ is chosen for this kernel function with d_0 being initial spacing between particles.

SPH approximations for the first-order and second-order gradients of function $f(\mathbf{x}_i)$ can be obtained by applying Taylor series expansion of function $f(\mathbf{x}_j)$ about $\mathbf{x}_i$. Among several formulations reported in the literature (Bui & Nguyen, 2021; Lian *et al.*, 2021), the following corrected SPH formulation for the first-order gradient of $f(\mathbf{x}_i)$ is adopted:

$$\widetilde{\nabla} f_i \approx \sum_{j=1}^{N} V_j \left(f_j - f_i\right) \widetilde{\nabla}_i W_{ij} \qquad \text{and} \qquad \widetilde{\nabla}_i W_{ij} = \mathbf{L}_{ij} \cdot \nabla_i W_{ij} \tag{33}$$

where $\mathbf{L}_{ij} = \left[\sum_{j=1}^{N} V_j \left(x_j - x_i\right)^m \nabla^n{}_i W_{ij}\right]^{-1}$ is the normalised matrix; and m and n indicate the coordinate direction with repeated indices implying summation.

On the other hand, there also exist several robust SPH formulations for the approximation of the second-order derivatives of a function (Bui & Nguyen, 2021; Cleary & Monaghan, 1999; Español & Revenga, 2003; Lian *et al.*, 2021; Chen et al. 2000). In this work, the general SPH approximation for the second-order derivatives, recently proposed by the authors and achieved excellent accuracy for highly disordered particle systems, is adopted and takes the form of (Lian *et al*., 2021):

$$\frac{\partial^2 f_i}{\partial x^m \partial x^n} = \sum_{j=1}^{N} V_j \left(f_j - f_i\right) \mathcal{D}^{mn} \tilde{F}_{ij} - \frac{\partial f_i}{\partial r^{m'}} \sum_{j=1}^{N} V_j r_{ji}^{m'} \mathcal{D}^{mn} \tilde{F}_{ij} \tag{34}$$

where $\mathcal{D}^{mn} = 4\frac{r_{ji}^m r_{ji}^n}{\left|\boldsymbol{r}_{ji}\right|^2} - \delta^{mn}$ with δ^{mn} being the Kronecker delta function; $\tilde{F}_{ij} = \frac{\mathbf{r}_{ji}}{\left|\mathbf{r}_{ji}\right|^2} \cdot \widetilde{\nabla}_i W_{ij}$ is the scalar part of the normalised kernel gradient with $\mathbf{r}_{ji} = \mathbf{x}_j - \mathbf{x}_i$ being the distance vector; and $\partial f/\partial r^{m'}$ is the first-order spatial gradient calculated from Eq.(33). It is noted that Eq. (34) shares a similar form to that proposed by Español & Revenga (2003), except the last term on the right-hand side of the equation, which is resulted from the Taylor series expansion and helps to remove approximation errors caused by particle distortion (Lian *et al.*, 2021). In this work, Eqs. (31), (33) and (34) are adopted to approximate the governing equations for the coupled seepage-deformation framework in the next section.

SPH discretisation of the governing equations

The mass balance equation for the solid phase in the unsaturated soil mixture governs the variation of the solid void fraction and thus the porosity. To solve this equation, one needs to convert it into the SPH approximation form, which can be achieved by applying Eq. (33) to the divergence term in Eq. (2), leading to the following SPH approximation for the porosity:

$$\frac{dn_i}{dt} = \sum_{j=1}^{N} V_j \left(1 - n_j\right) \mathbf{v}_{ji}^s \, \widetilde{\nabla}_i W_{ij} \tag{35}$$

where $\mathbf{v}_{ji}^s = \mathbf{v}_j^s - \mathbf{v}_i^s$ is the relative velocity between two particles. Once the porosity is determined from the above equation, the variation of saturated permeability and the specific moisture term can be updated.

Similarly, the SPH approximation for the seepage flow equation can be obtained by applying Eqs. (32)-(33) to discretise the spatial gradient terms in Eq. (14), leading to the following SPH approximation for the evolution of the water pressure head:

$$\begin{aligned}\frac{dh_i}{dt} = \frac{1}{\tilde{C}_{Sr}} \Bigg\{ & \sum_{j=1}^{N} V_j \bar{k}_{mn}^{ji} H_{ji} \mathcal{D}^{mn} \widetilde{F}_{ij} - k_i^{mn} \frac{\partial H_i}{\partial r^{m'}} \sum_{j=1}^{N} V_j r_{ji}^{m'} \mathcal{D}^{mn} \, \tilde{F}_{ij} \\ & - S_r^i \sum_{j=1}^{N} V_j \mathbf{v}_{ji}^s \, \widetilde{\nabla}_i W_{ij} + \frac{1}{g} \sum_{j=1}^{N} V_j \bar{k}_{mn}^{ji} \left(\frac{d\mathbf{v}_j^s}{dt} - \frac{d\mathbf{v}_i^s}{dt} \right) \widetilde{\nabla}_i W_{ij} \Bigg\}\end{aligned} \tag{36}$$

where $\bar{k}_{mn}^{ji} = \left(k_{mn}^i + k_{mn}^j\right)/2$ is the arithmetically mean value of water permeability; and $H_i = (h_i + z_i)$ is the total water pressure head.

Finally, once the water pressure head (h) is determined, the degree of saturation, pore-water pressure and total stress can be determined. This information can be subsequently used to update the motion of solid particles by solving the momentum equation Eq. (11), which can also be converted into the SPH approximation form as follows:

$$\left(\frac{d\mathbf{v}_s}{dt}\right)_{\boldsymbol{i}} = \sum_{j=1}^{N} m_j \left(\frac{\boldsymbol{\sigma}_j}{\rho_j^2} + \frac{\boldsymbol{\sigma}_i}{\rho_i^2} \right) \nabla_i W_{ij} + \mathbf{C}_{ij} + \mathbf{b} \tag{37}$$

where $\mathbf{C}_{ij}$ is a stabilisation term, which consists of the artificial viscosity, artificial stress and damping force, utilised to remove the stress fluctuation and tensile instability in SPH. Readers are referred to previous works (Bui *et al.*, 2008, 2011a, 2011b; Nguyen *et al.*, 2017; Bui & Nguyen, 2021) for detailed formulations and selections of required parameters for these terms.

Time integration

In this study, the Leap-Frog (LF) algorithm is used to integrate the SPH approximation equations developed in the preceding section. In this approach, the state variables (Υ), including porosity (n), water pressure head (h), effective stress ($\boldsymbol{\sigma}$') and velocity ($\mathbf{v}$), are updated at mid-step in time, while the displacement vector is updated at the full-time step as follows:

$$\Upsilon_{t+\Delta t/2} = \Upsilon_{t-\Delta t/2} + \Delta t \left(\frac{d\Upsilon}{dt}\right)_t$$
$$\boldsymbol{x}^s_{t+\Delta t} = \boldsymbol{x}^s_t + \Delta t \cdot (\mathbf{v}^s)_{t+\Delta t/2} \tag{38}$$

The stability of the LF time integration scheme is maintained by the Courant-Friedrichs-Lewy (CFL) condition. For the mechanical calculation, the timestep is restricted by:

$$\Delta t_s \le C_{CFL} \frac{h_{sml}}{c_{sp}} \tag{39}$$

where C_{CFL} is a constant, which is taken to be 0.1 throughout this paper; $c_{sp} = \sqrt{E_s/\rho_s}$ is the speed of sound in the solid phase with E_s being Young's modulus of the solid phase.

On the other hand, for the hydraulic calculation of the seepage flow, the timestep is restricted by the following condition (Lian *et al.*, 2021):

$$\Delta t_l \le C_{CFL} \frac{\tilde{C}_{Sr} h^2_{sml}}{k^{sat}} \tag{40}$$

Therefore, the overall timestep requires for both analyses is:

$$\Delta t \le \min(\Delta t_s, \Delta t_l) \tag{41}$$

The issue associated with the condition (41) is that the mechanical timestep Δt_s is often several orders of magnitudes smaller than the hydraulic timestep Δt_l. Thus, if one adopts Δt_s for a coupled analysis of a long-term diffusive process, an extremely high computational cost would be required and, in many cases, are not necessarily essential when the deformation of the solid phase is negligible. To address this problem, an adaptive two-timestep coupling technique is proposed and will be presented in the next section.

An adaptive time-stepping scheme for coupled flow-deformation analysis

As discussed in the preceding section, to reduce the overall computational cost required for coupled analysis of long-term diffusive problems, such as rainfall-induced slope failure or seepage flow through low permeable materials, an adaptive time-stepping scheme is proposed to enable the use of different timestep sizes for mechanical and hydraulic analyses. Figure 2 shows the schematic diagram of the proposed time integration scheme. The mechanical timestep (Δt_s) will strictly follow condition (39) and is used to update the mechanical variables such as effective stresses and soil displacements. On the other hand, depending on the magnitude of deformation, fully coupled or loosely coupled schemes can be adopted. In particular, when the soil undergoes large deformation, to properly describe the coupled flow-deformation behaviour, the fully coupled scheme is adopted and this requires $\Delta t_l = \Delta t_s$, which strictly satisfies the condition (41). In contrast, when the soil undergoes negligible deformation, the loosely coupled scheme can be adopted and the timestep Δt_l required for the hydraulic computation could be defined as follows:

$$\Delta t_l^D = F\Delta t_s = \max\left\{\frac{\tilde{C}_{sr} h_{sml}}{c_{sp} k^{sat}} \exp\left(-D\epsilon_{eq}^p\right), 1\right\} \Delta t_s \tag{42}$$

where ϵ_{eq}^p is the accumulated equivalent plastic shear strain for the entire computation (see ***Appendix A***); D is a parameter governing the rate of change of the adaptive hydraulic time-step or how fast the adapting hydraulic timestep is approaching the mechanical one; $F = \max\left\{\frac{\Delta t_l}{\Delta t_s} \exp\left(-D\epsilon_{eq}^p\right), 1\right\}$ defines the maximum number of increments that the fluid timestep can be increased from the solid timestep.

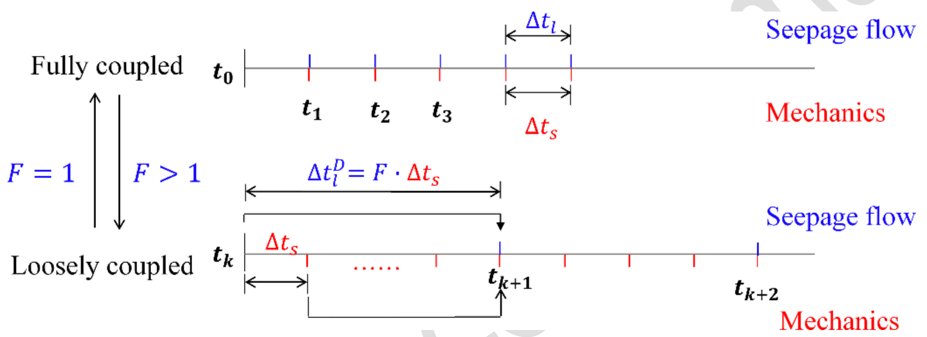

Fig. 2. Schematic diagram of the adaptive two-timestep scheme for the coupled flow-deformation analysis

Equation (42) indicates that the accumulated equivalent plastic shear strain plays a crucial role in controlling the adaptive hydraulic timestep. When the soil undergoes negligible plastic shear deformation (i.e. $\epsilon_{eq}^p \approx 0$), the influence of soil deformation on the seepage flow is also negligible. Thus, the adaptive hydraulic timestep Δt_l^D reaches its maximum value set by condition (40). On the other hand, as the soil undergoes large plastic shear deformation, ϵ_{eq}^p increases, causing Δt_l^D to reduce. During this process, there is a gradual transition from the loosely coupled scheme to the fully coupled scheme, as shown in Fig. 2, where Δt_l^D gradually approaches Δt_s as F reduces to 1. To be more specific, for a general case where $\Delta t_l^D \gg \Delta t_s$ and within a single SPH computational cycle of Δt_l^D, the mechanical variables are only updated once after each Δt_l^D using the timestep Δt_s. All mechanical parameters are subsequently kept unchanged during the rest of Δt_l^D. This process is then repeated for the next increment of Δt_l^D, in which the updated mechanical parameters obtained from the previous Δt_l^D increment are passed to the next Δt_l^D increment (i.e. beginning of the next increment) to update hydraulic variables.

The above adaptive time-stepping technique offers a superior way to balance the accuracy and computational cost for SPH simulations with coupled flow-deformation problems. The suitability of this approach will be investigated when applying this technique for studying the rainfall-induced slope failure experiment.

VERIFICATIONS AND APPLICATIONS

1-D Terzaghi's consolidation problem

The proposed SPH framework is first validated against the 1-D Terzhaghi consolidation problem, whose theoretical solutions are available (see ***Appendix B***). In this test, a fully saturated soil column of 1.0 m high and 0.1 m width is modelled using 1000 particles with an initial spacing distance of 0.01 m, as shown in Fig. 3a. The boundary conditions are set to be smooth on both sides and fully fixed at the bottom (Bui, Fukagawa, *et al.*, 2008). The soil is modelled by an isotropic linear elastic material, whose properties are listed in Table 1. Initially, the effective stress and excess pore-water pressure are set to zero, while the gravity is disregarded in this case. The consolidation test is conducted in two stages, i.e., the loading state and the dissipation stage. During the loading stage, a ramp-load was applied to the upper surface (see Fig. 3b) by imposing the following vertical acceleration to the top surface particles:

$$a_y = \begin{cases} -\dfrac{q_0}{dx \cdot [(1-n)\rho_S + n\rho_l]}\dfrac{t}{t_L} & t \leq t_L \\ -\dfrac{q_0}{dx \cdot [(1-n)\rho_S + n\rho_l]} & t > t_L \end{cases} \tag{43}$$

where dx is the particle spacing distance; $q_0 = 10\,kPa$ is the surcharge load; t is the computational time; and $t_L = 0.01\,s$ is the loading time. Furthermore, to suppress the stress fluctuation caused by the applied load, damping force with the non-dimensional damping coefficient $\xi = 0.04$ is adopted (Bui & Fukagawa, 2013). During the loading stage, all boundaries are kept impervious to ensure no pressure dissipation. The development of the excess pore-water pressure is illustrated in Fig. 3c, which indicates that all the applied load is transferred to the excess pore-water pressure at the end of the loading process. Thereafter, a drainage condition (i.e. $p_l = 0$) is imposed on the top surface to trigger the dissipation process. Figure 4 shows a comparison between the SPH and theoretical solutions for the variation of excess pore-water pressure within the soil column during the dissipation process. It can be seen that an excellent agreement between the proposed SPH model and theoretical solutions was achieved. Along with the pore-water pressure dissipation, the development of effective stress within the soil column, as shown in Fig. 5, at three particular locations, also matches well with the analytical solutions. Furthermore, the surface settlement predicted by the newly developed coupled SPH model again showed excellent agreement with the analytical solution, as illustrated in Fig. 6. These excellent agreements suggest that the proposed SPH framework could capture well the coupled behaviour of saturated soils and is ready for more complex coupled hydro-mechanical problems, which will be presented in the next section.

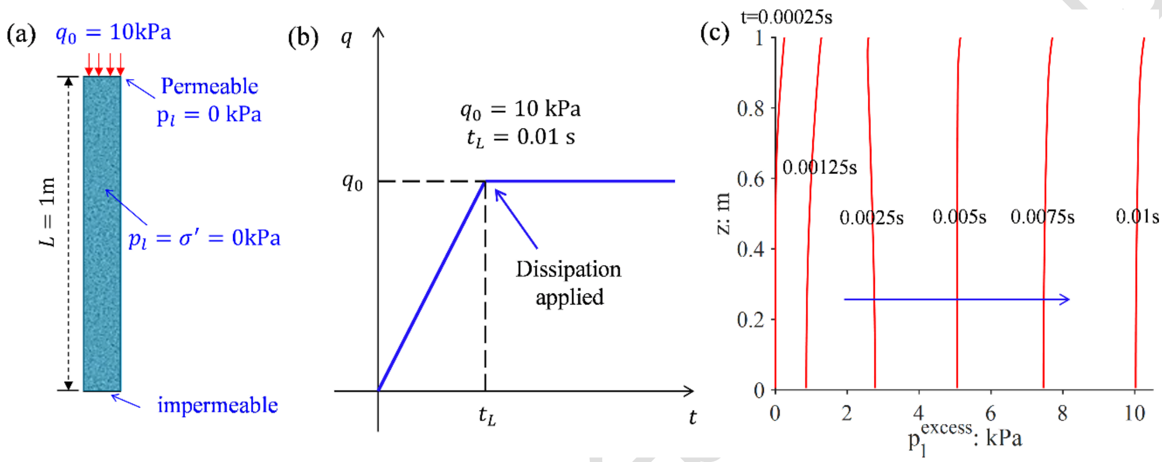

Fig. 3. Model test of 1D consolidation: (a) Geometry and boundary conditions;(b) Ramp load applied on the soil column; (c) Evolution of the excess pore water pressure

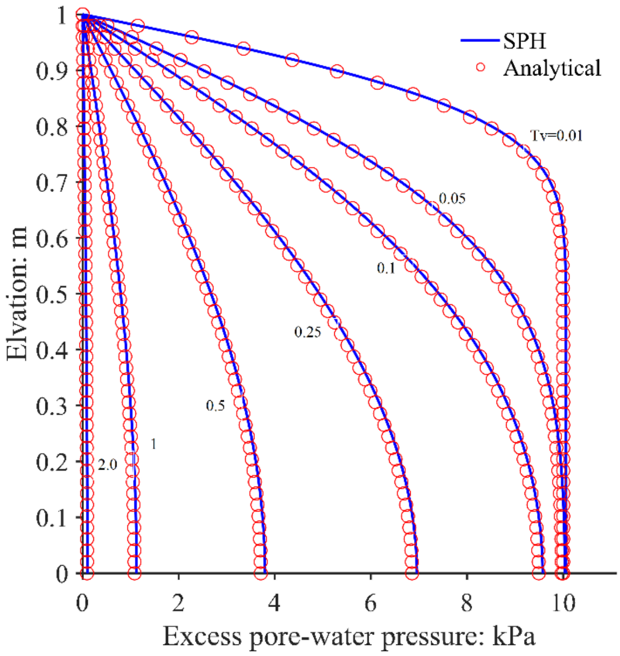

Fig. 4. The evolution of the excess pore-water pressure

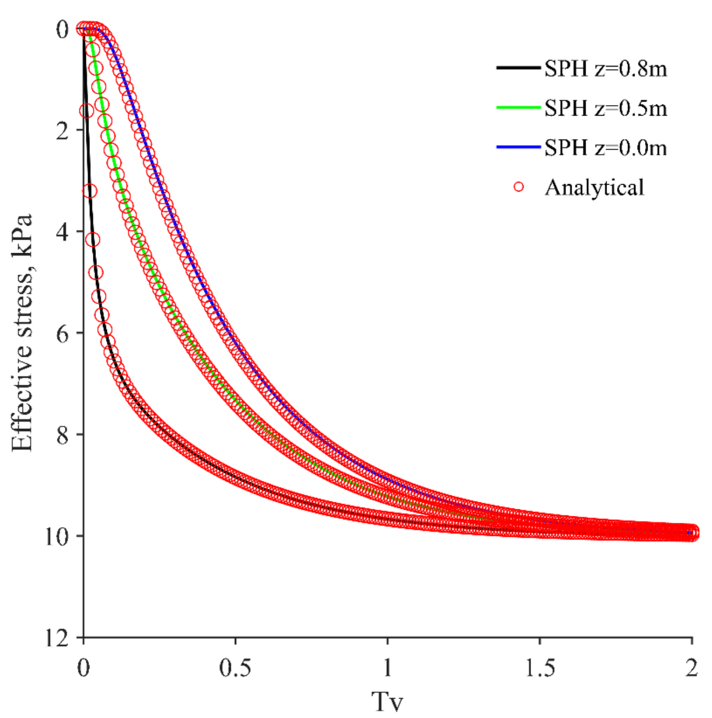

Fig. 5. The evolution of the effective stress at various elevations.

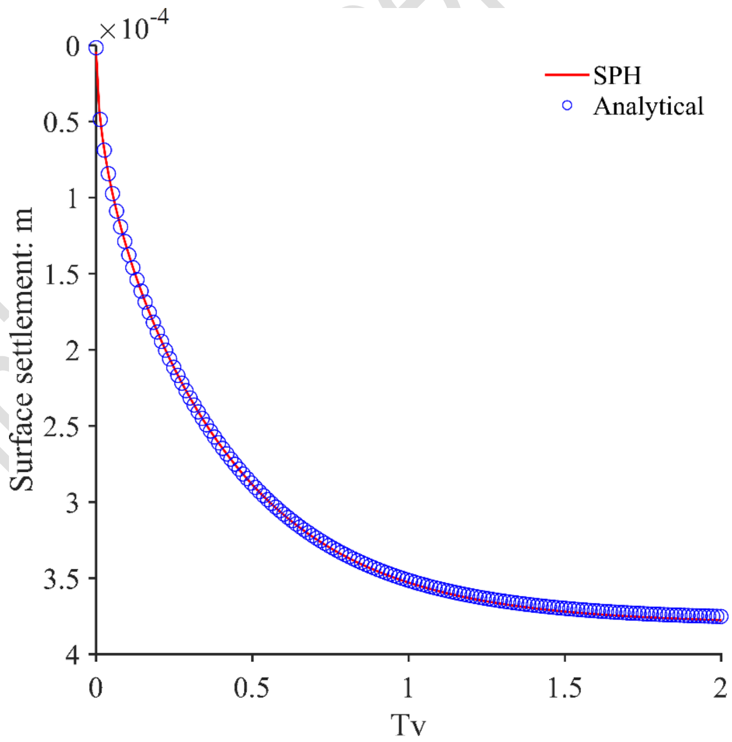

Fig. 6. The evolution of the surface settlement

Modelling rainfall-induced slope failure experiment

Case description and model setup

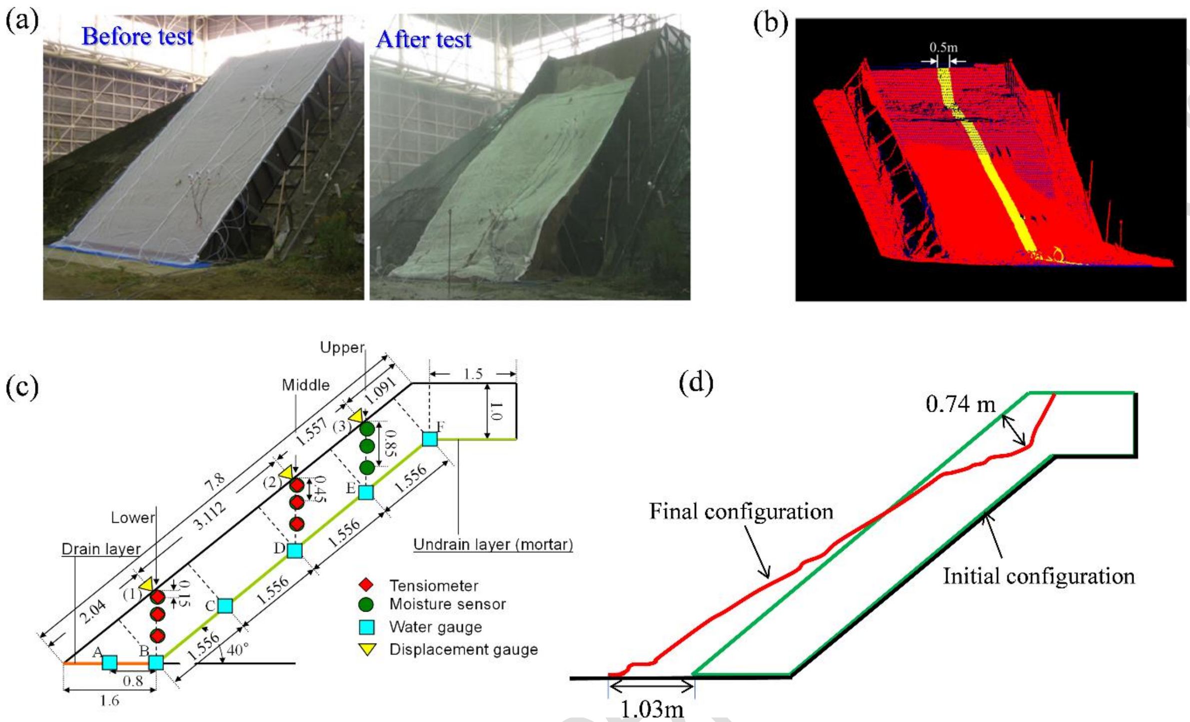

Fig. 7. The slope configuration in the experiment: (a) View of the slope before and after the infiltration test; (b) 3D scanned data of the final configuration; (c) Schematic diagram of the experimental setup; (d) The scanned configurations of the slope before and after failure.

The case studied here is inspired by a large-scale rainfall-induced slope failure experiment previously reported by Danjo *et al.* (2012; 2013; 2015). Figure 7 outlines the experimental setup and testing results. A shallow slope of 4.0 m width, 5.0 m height, and 7.0 m length was constructed on a non-deformable steel frame, as shown in Fig.7a. The main body of this model slope was made of unsaturated Masa sand (i.e. a weathered granite soil) and formed a slope angle of 40° to the horizontal surface (Danjo *et al.*, 2012; 2013; 2015). Six tensiometers were installed at depths of 0.15 m, 0.45 m, and 0.85 m in the lower and middle sections of the slope, marked diamond symbols in Fig. 7c, to measure the evolution of pore-water pressure within the slope. Several water gauges were also installed at the bottom to measure the increase in groundwater caused by rainfall infiltrated water. Furthermore, to measure the surface displacement of the slope, three displacement gauges were also installed on the slope surface, marked by the triangle symbol in Fig. 7c (Danjo *et al.*, 2012). To simulate the rainfall condition, an artificial rainfall machine was used to generate water flux on the slope surface until the slope failure occurred. After the slope collapse, a 3D laser scanner was used to obtain the surface morphology and final configuration of the slope, which are presented in Fig. 7b and Fig. 7d, respectively. All other relevant experimental data were reported in Danjo *et al.* (2012; 2013; 2015) and are used to assess the performance of the proposed coupled SPH model and will be later presented together with our predicted results.

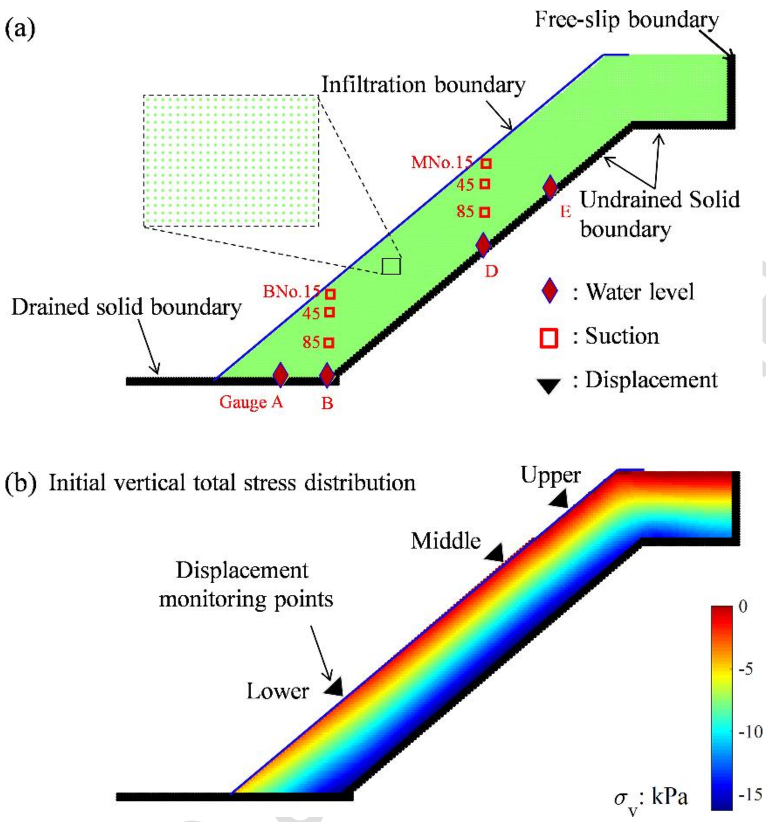

Fig. 8. Model setup: (a) The initial particle configuration and boundary conditions of the SPH model; (b) Initial in-situ vertical total stress distribution in the slope model

A total of 5911 particles with an initial spatial distancing of 0.04 m are used to generate the above slope model, as shown in Fig. 8a. The rigid frame is modelled using five layers of virtual particles with either the fully-fixed or the free-slip boundary (see Fig. 8a). Both types of boundaries can be modelled using the standard approaches reported in (Bui, Fukagawa, *et al.*, 2008). All hydraulic boundary conditions are modelled following the work recently proposed by the authors (Lian *et al.*, 2021). It is noted that, although a drainage layer was installed at the slope base in the experiment (Fig.7c), the water gauges installed at A and B recorded a small water accumulation amount. To reflect this condition in the experiment, an experimentally calibrated "ponding" condition of $p_l \leq 0.25kPa$ is imposed at the slope base. In addition, a prescribed negative pore-water pressure ranging from -2.75 to -2.95 kPa is linearly distributed along the slope elevation to generate an initially unsaturated soil condition, reflecting the initial suction values measured at tensiometers. Before the water infiltration is applied, an initial stress field is obtained by applying gravitational load to the slope model, as shown in Fig. 8b. Thereafter, a constant negative pore-water pressure of -0.715 kPa (i.e. experimentally recorded values from tensiometers) is imposed on the slope surface to model the rainfall infiltration and is maintained on the ground surface during the entire computational process. Due to hydraulic gradient differences, a rainfall-induced infiltrated seepage flow is achieved in the slope.

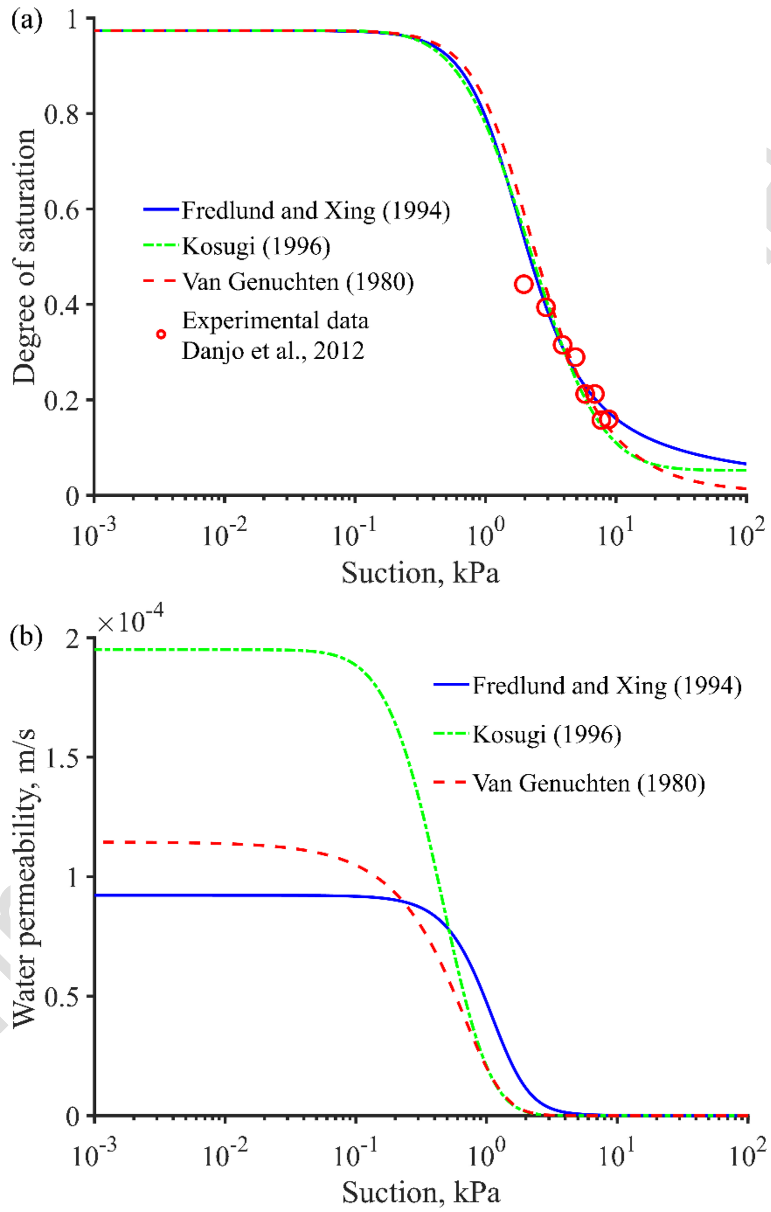

Fig. 9. Hydraulic constitutive models adopted in the calculation: (a) Fitting of soil water characteristic curves (SWCS) to the experimental data; (b) Water permeability curves.

On the other hand, the mechanical parameters of the soil slope were not reported in the series of studies by Danjo *et al.* (2012; 2013; 2015). Therefore, these parameters were extracted from the literature data reported on Masa soil (Kansai area, Japan) and are summarised in Table 2, some of which are calibrated for the suction-dependent elastoplastic DP softening model. Besides, to describe the link between suction and degree of saturation, three well-known Soil Water Retention Curves (SWRC) (Fredlund & Xing,1994; van Genuchten,1980; Kosugi,1996) are considered in this study, as shown in Fig. 9. Our numerical tests confirmed that these hydraulic constitutive models provide similar results in predicting the rainfall-induced seepage flow behaviour. Accordingly, only results predicted by van Genutchen's SWRC are presented in this study, and the corresponding hydraulic constitutive parameters are listed in Table 2. Furthermore, to reflect the anisotropic characteristics of seepage flow, in which the horizontal seepage is expected to be larger than the vertical one, this study assumes that $k_x = 1.2k_{sat}$ and $k_y = 0.8k_{sat}$. These parameters are again calibrated to best fit experimental data. Finally, it is noted that this rainfall-induced slope failure simulation involved a long-term seepage infiltration process (i.e., 167 minutes), but the slope failure process only lasted a few seconds (Danjo *et al.*, 2012; 2013; 2015). Thus, if a minimum timestep size satisfied condition (41) is applied, it would take a few days to model such a long physical test. To address this issue, the proposed adaptive time-stepping integration scheme is adopted. The mechanical timestep is $\Delta t_s = 0.0003$s, while the maximum hydraulic timestep is $\Delta t_l = 0.05$s, corresponding to a controlling parameter of $D = 5$. Finally, it is worth noting that, to accelerate the computational speed, we have deliberately reduced the water bulk modulus used in this simulation, and this enables the adoption of a larger hydraulic timestep according to the CFL condition (40). We have tested our simulations for larger values of the water bulk modulus and did not find any significant difference in the numerical results.

Qualitative analysis of slope response to rainfall

The attention is first paid to the predicted rainfall-induced seepage infiltration and progressive failure of a slope, as shown in Fig 10. The imposed rainfall, in the form of constant water pressure, on the ground surface causes water flux to infiltrate into the soil. As the wetting front is processing in the slope, an increase in the negative pore-water pressure from around -3.0 kPa to -0.72 kPa is observed within the wetting zone (T ≈ 44.41 min). During this stage, the slope deformation is negligible and no plastic deformation occurs. This suggests that the proposed adaptive time-stepping scheme has helped to significantly reduce the overall computational cost. At around 145.92 min, the seepage flow reaches the rigid frame at the bottom slope and is subsequently trapped, forming a fully saturated region associated with positive pore-water pressure. At the same time, the plastic strain in the slope starts to develop at the slope base and gradually propagates to the upper part of the slope crest, forming a well-defined failure surface. The development of plastic strain in the slope is attributed to the reduction of soil suction caused by the increase in the degree of saturation. From this stage onward, as deformation develops, the hydraulic timestep is adaptively reduced and eventually reaches its minimum value corresponding to the mechanical timestep size. As the entrapped water content continues accumulating at the base of the slope and migrates downwards to the slope toe, an elliptic fully-saturated region is observed in the slope (see Fig. 10 with T = 165.51 min). At the same time, the groundwater level rises significantly, further reducing the soil suction and facilitating the slope failure process. As a result, a clear localised shear band (or sliding surface) is well developed, connecting the toe and the crest of the slope, indicating an acceleration of the slope failure. All of a sudden, a quick movement of the mobilised volume occurs and slides rapidly along the non-deformable frame bottom at T = 166.03 min, leading to a detachment of the sliding body from the intact scarp on the upper supporting frame. Within the shear band, it is also interesting to see how the pore-water pressure changes are caused by the volumetric deformation (Fig. 10). Because of the soil dilation, volumetric expansion is expected to occur as the soil undergoes shearing, leading to a reduction of positive pore water pressure (T = 166.03 min), all of which can be well captured, thanks to the proposed fully coupled SPH framework. The landslide stops at 166.06 min, but the entire sliding process lasts for only 0.04 min, which is a significantly small timescale compared with that of the whole rainfall infiltration process. Even with such a significant difference in the timescale, the dynamic coupling process of both seepage flow and soil deformation can be still effectively captured, thanks to the newly proposed adapting time-stepping algorithm. To further quantitatively verify the capability of the proposed fully coupled SPH framework in predicting the seepage infiltration process and the failure response of the slope, the calculated evolutions of suction, water level, and ground displacement are assessed against experimental data.

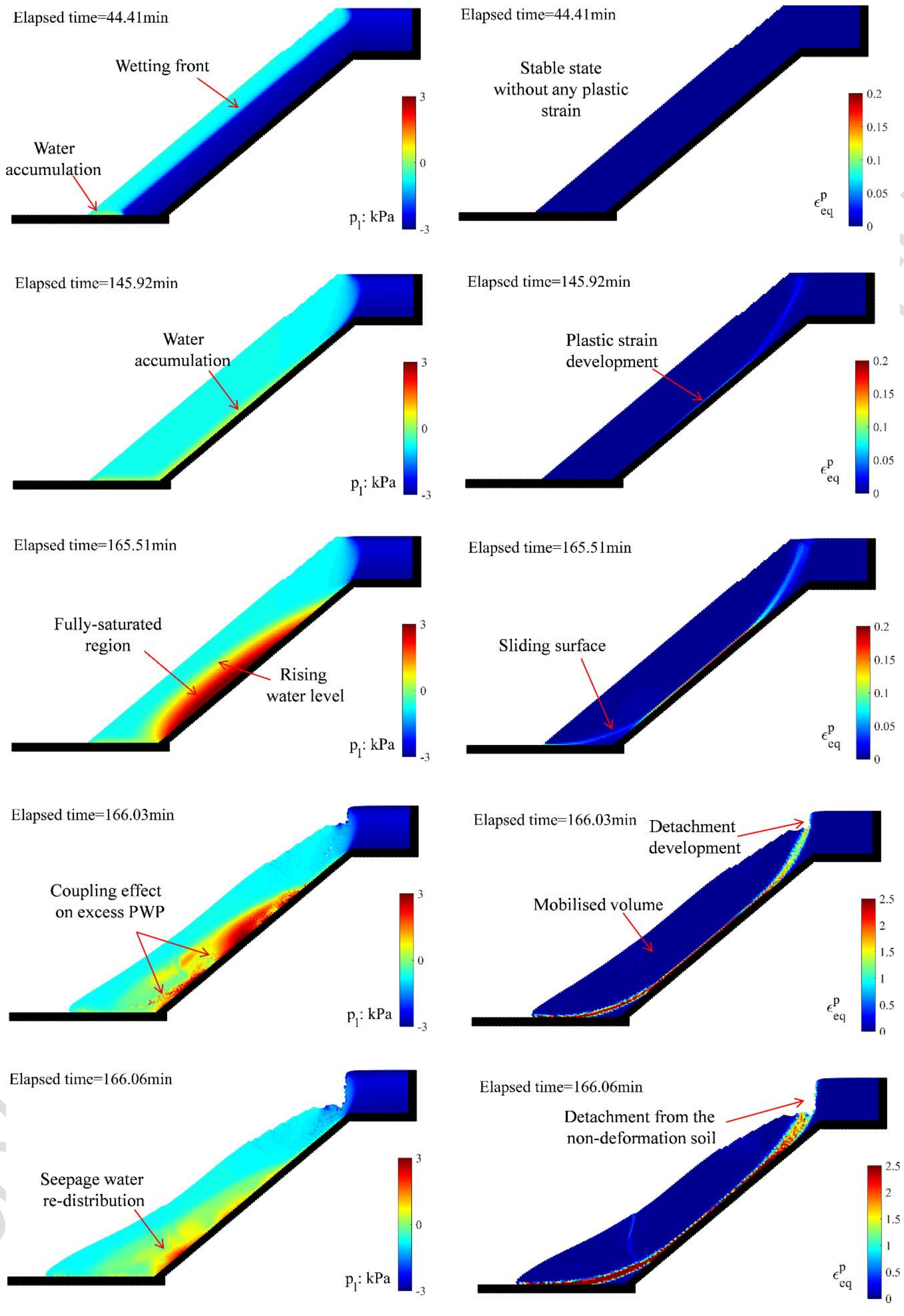

Fig. 10. SPH predictions of the evolutions of pore-water pressure and equivalent plastic strain at different time intervals.

Quantitative analysis of seepage infiltration into the slope

Figure 11 shows a comparison between the SPH prediction and experiment for the evolution of suction captured from 6 monitoring points due to the rainfall infiltration. A fairly good agreement between the predicted results and experimental data is achieved across all monitoring points, suggesting that the newly developed SPH model could capture well the unsaturated seepage flow through deformable media. For example, in the middle section of the slope (i.e. MNo.15, MNo.45 & MNo. 85), suction starts to decrease at MNo.15 nearly 10 min after the rainfall begins and gradually reaches its temporary equilibrium state at around 30 min, corresponding to negative pore-water pressure around 0.715 kPa. With further progression of the wetting front under the gravity, suction at the other two measuring points (i.e. MNo.45 and MNo.85) follow the same trends, which react to the wetting process and reach their temporary equilibrium state at further delay times because these monitoring points are located at deeper locations from the slope surface. During the temporary equilibrium state, the soil at all monitoring points is still under the unsaturated condition, which is evidenced by the negative PWP value at all points. As the infiltration seepage flow continues, seepage water reaches the impervious bottom boundary and gradually accumulates, forcing a transition from partially saturated to fully saturated conditions and raising the groundwater level inside the slope. At around 150 min, this rising groundwater level is first captured at MN0.85, of which the suction further decreases to nearly zero (i.e., saturated). A similar evolution of suction also can be seen in other measuring points installed close to the slope toe (i.e. BNo.15, BNo.45 & BNo.85). It is worth noting that there still exist some noticeable differences between the results predicted by SPH and those in the experiment. For example, a faster seepage flow process is observed in SPH (i.e., BNo.85 in Fig.11a). This can be attributed to the existence of air bubbles in the pores of the soil in the experiment, which would delay the penetration of the seepage flow. However, this delay effect caused by air bubbles was not considered in this study. Another possibility could be due to the highly anisotropic and random properties of the soil in the experiment, which were also not considered in the present numerical model. Nevertheless, a reasonably good agreement in the infiltration process and water level development can be seen, showing the effectiveness of the proposed framework.

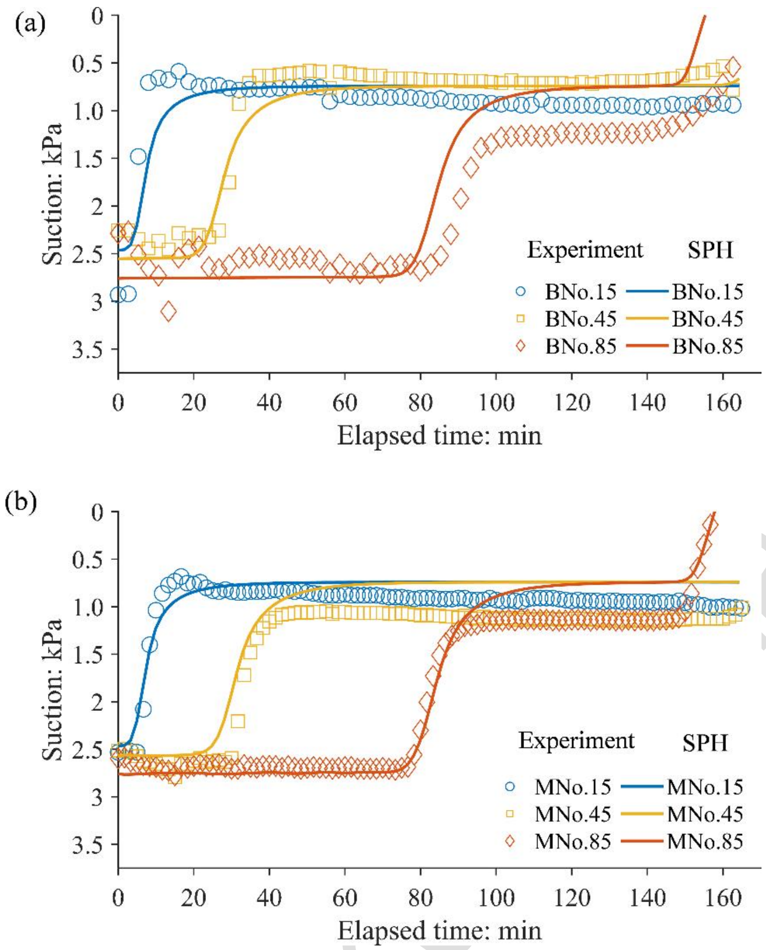

Fig. 11. Comparison between the SPH predicted results and the experimental data for the evolution of the active pore-water pressure (PWP) at several monitoring points in (a) the middle section and (b) the lower section of the slope.

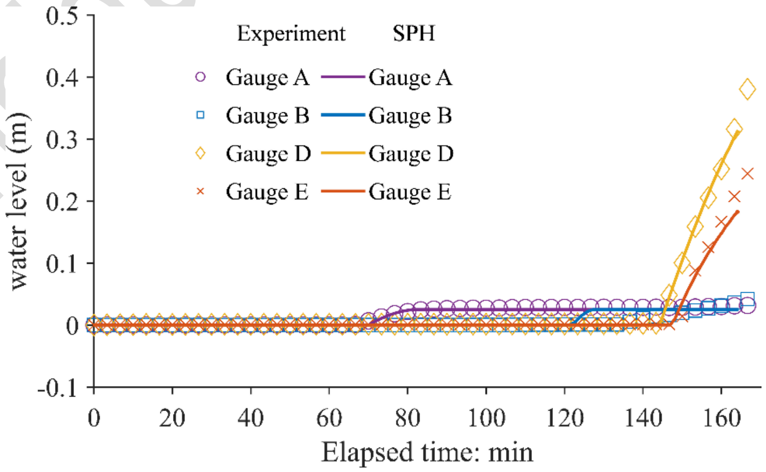

Fig. 12. Comparison between the SPH predicted results and the experimental data for the water level at the slope base

The predicted water level at the base of the slope is compared with the experimental data and shown in Fig.12. Again, very good agreements with the experiment are achieved by the proposed SPH framework. For example, it is observed in both SPH and experiment that Gauges A and B respond to the water level change at around 65 min and 100 min, respectively. After that, the water level at these locations gradually increases and reaches an equilibrium state, corresponding to the “ponding” condition (i.e. 0.25 kPa). On the other hand, the water level at both Gauges D and E increases rapidly at around 143~145 min when the infiltrated water flow reaches the undrained bottom boundary and accumulates. The predicted water level at these two locations also agrees well with the experimental counterparts, suggesting that the proposed SPH framework could capture well the transition of seepage flow from the unsaturated to fully saturated conditions.

Quantitative analysis of the failure response of the slope

Ground surface displacement monitoring is one of the most important measures in slope failure forecast and prevention systems. Therefore, it is crucial to develop a robust computational approach capable of correctly predicting this information. In this section, the evolution of the ground displacement predicted by the proposed SPH model is quantitatively analysed to demonstrate its applicability for future predictions of rainfall-induced landslides. Fig. 13 shows a comparison between the numerical prediction and the experiment for the surface displacement during rainfall loading. It can be seen that very good agreements between the predicted results and experimental data are achieved, suggesting that the proposed SPH framework could capture well the time-dependent deformation response of the slope caused by rainfall infiltration. For example, the slope remains essentially stable during the first 140 min, thanks to the contribution of suction to the overall shear strength of the soil slope. However, from around 140 min onward, a bending upward trend in the displacement curves develops, indicating that the surface displacement starts to occur as the groundwater level grows (see Fig.12). This gradually-developed displacement prior to the quick sliding of the slope can be attributed to the increase in the soil mass caused by increases in the degree of saturation and rainwater accumulation inside the soil body. The minor slope deformation before its collapse is also known as the precursor state of slope failure (Moriwaki, 2001; Moriwaki *et al.*, 2004). As the groundwater level rises, the surface movement accelerates, and the slope moves into the warning state with a noticeable sliding trend. This acceleration of soil movement is mainly caused by the rising water level, leading to the loss of the suction-dependent shear strength of the soil slope. As a result, the slope cannot maintain its stable condition and thus suddenly collapses at around 166 min, indicated by a rapid upward (non-converged) trend in the displacement curve in Fig.13. The timing of slope failure (or failure state) predicted by the SPH model is nearly equivalent to the experimentally observed counterpart. Nevertheless, the SPH model underestimated the surface displacement during the warning state, and this difference could be attributed to the suitability of constitutive models. In particular, the suction-dependent constitutive model adopted in the current SPH framework is relatively simple and thus may not be able to fully capture the unsaturated soil behaviour in the experiment. To address this issue, advanced unsaturated soil constitutive models should be adopted, along with tests to provide sufficient data for calibrating and validating those models, all of which are out of the scope of this paper.

Fig. 14 shows the progressive deformation patterns caused by the rainfall infiltration, visualised through the deformed grids of 'tracers' plotted on the slope body. The entire slope collapsing process lasts for a short period. At around T = 166 min, a noticeable settlement is observed on the crest of the slope, where minor shear deformation can also be noticed in the zoom-in figure. At this stage, no deformation occurs at the lower portion of the slope. After that, the soil deformation progresses rapidly, and at around T = 166.03 min, the mobilised soil volume detaches from the upper non-deformable scarp. At this point, severe shear deformation is observed along with the sliding surface. The sliding mass decelerates and stops at around T = 166.05 min, and the predicted final failure configuration matches well with the longitudinal cross-section of the slope obtained by the 3D laser scanner in the experiment (Danjo *et al.*, 2012; 2013; 2015). The final run-off distance predicted by SPH is about 0.95 m, which agrees well with the reported experimental data of 1.03 m. At the same time, the largest settlement of the slope surface predicted by SPH is around 0.82 m depth, which is also close to the experimental data of 0.74 m. Therefore, it can be concluded that the proposed coupled flow-deformation SPH model could predict well the large deformation and failure responses of an unsaturated soil slope caused by rainfall infiltration.

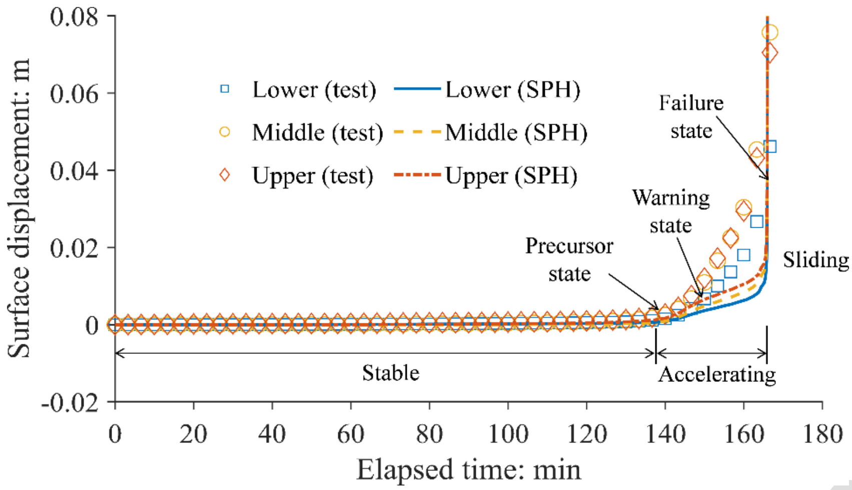

Fig. 13. Comparison between the SPH predicted result and the experimental data for the surface displacement against time.

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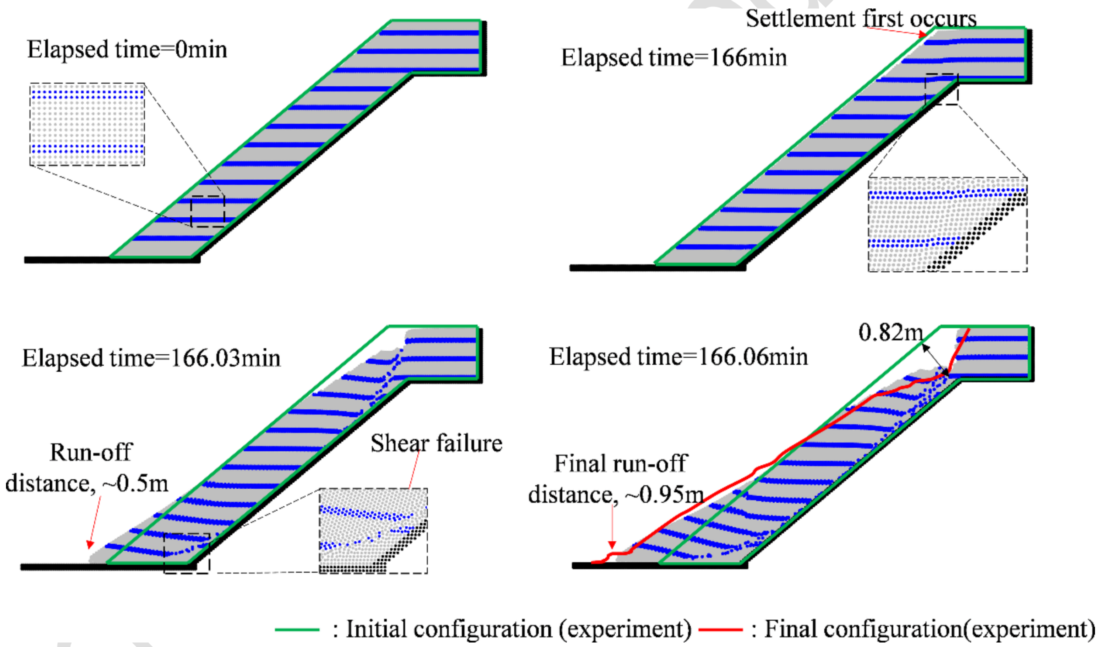

Fig. 14. Progressive deformation patterns of the slope predicted by SPH and its comparison against the experiment for the final failure configuration.

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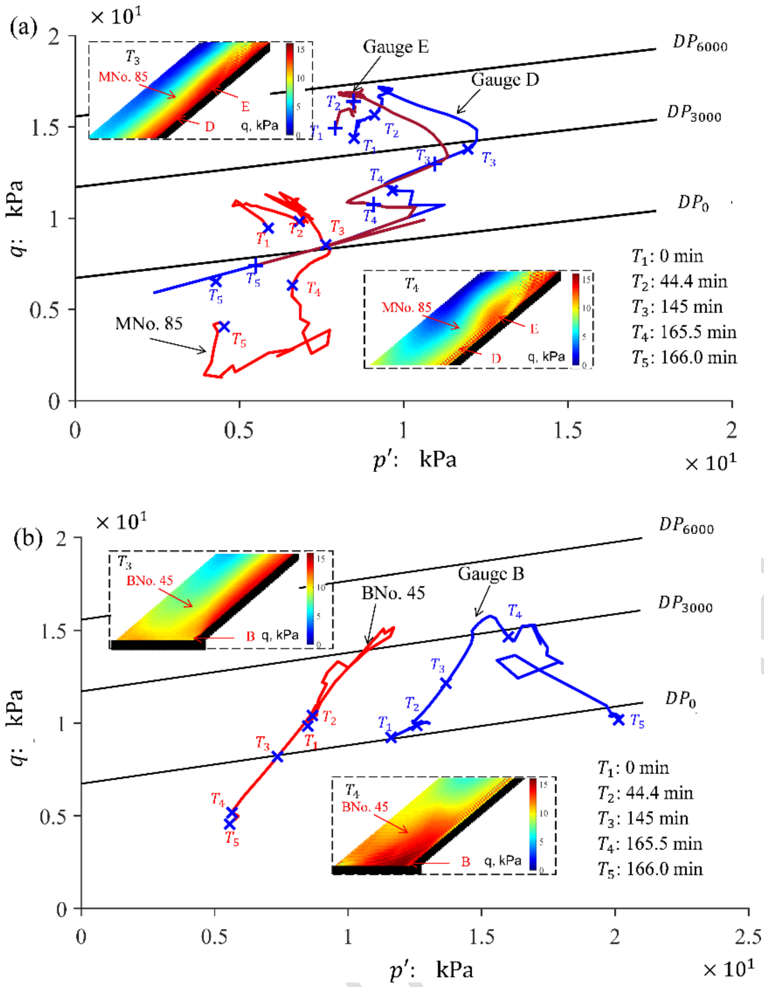

Fig. 15. Calculated evolution of effective stress at several monitoring points: (a) MNo. 85, Gauge D and E; (b) BNo. 45 and Gauge B.

Stress path analysis

The capability of the proposed model to provide insights into stress variations inside a deformable unsaturated porous medium is demonstrated in this section. The stress at several monitoring points in Fig. 8a, corresponding to the locations inside (i.e. Gauges D and E) and outside (i.e. B, MNo. 85 and BNo. 45) the shear band, respectively, was monitored during the simulation. The stress paths in the deviatoric stress and effective mean stress ($p' - q$) plane are plotted in Fig. 15, where the Drucker-Prager yield surfaces corresponding to different suctions are also presented in the figure. During the first 45 min of rainfall loading (i.e. from T1 to T2), the effective mean stress at all locations increases, which can be attributed to the increase of soil mass due to wetting. Meanwhile, the shear stress at Gauges D and E is much higher than that at MNo. 85, indicating the precursor of shear band localisation at Gauges D and E. From T2 to T3, the reduction in soil suction due to water seepage causes the shear stress to reduce, while the mean effective stress at these locations continues to increase due to the increase of soil mass. From T3 to T4, due to the water accumulation and positive pore-water pressure generation, the effective mean stress at Gauges D/E and MNo. 85 starts to decrease. The shear stress at these locations also decreases, indicating that the materials at these two points undergo shear strength reduction associated with suction-induced softening. On the other hand, from T1 to T3, some noticeable fluctuations are observed in the stress paths, which could be attributed to the sudden change in the soil density caused by wetting and the dynamic nature of SPH modelling. The slide motion of the slope occurs at around T3, causing a sharp decrease in the effective shear stress at all Gauges D & E and MNo.85, which indicates that the soil at these locations undergoes further softening behaviour associated with increasing plastic shear deformation. The effective shear stress of particles located within the shear band (i.e. Gauges D & E) is found to be larger than those outside this zone (i.e. MNo.85), as shown in the contour plots of deviatoric stress in Fig. 15(a), indicating that particles within the shear band undergo higher shear deformation. The difference in the effective shear stress of particles located inside and outside the shear band could also be explained by the fact that they are located at different locations and thus are subjected to different stress loading paths and different pore-water pressure developments. Gauges D and E share nearly the same stress path since they are both located inside the shear band. The final stress state at all measuring points is located below the DP_0 line, which defines the peak shear strength envelop of the soil at the fully saturated condition, and this can be attributed to the softening behaviour of soils.

The evolution of the stress path at BNo.45 and Gauge B monitoring points located at the slope toe, in general, shows similar responses to the rainfall infiltration process compared to those monitoring points on the upper portion of the slope. However, different from the stress at Gauges D/E, the stress loading path at Gauge B, located below the shear band, is mainly characterised by increasing mean stress, which can be attributed to the increase in compression caused by the sliding mass from T3 to T5. The evolution of shear stress at Gauge B also reflects the suction-dependent and strain-softening constitutive behaviour adopted in the model, evidenced by the increase and subsequent reduction of the shear stress. Outside the shear band at the slope toe, it is interesting to notice that the stress loading path at BNo.45 monitoring point follows more and less the same loading path, undergoing loading and subsequent unloading process (i.e. T1 to T5). Compared to the stress loading paths at Gauges D/E, it appears that the shear stress inside the shear band undergoes continuous loading during the shearing process (i.e. from T1 to close to T4), while the shear stress outside the shear band undergoes loading and unloading. This observation of the stress loading path inside and outside the localisation zone is well supported by the double-scale constitutive theory (Bui & Nguyen, 2021; Nguyen & Bui, 2020; Nguyen *et al.*, 2016).

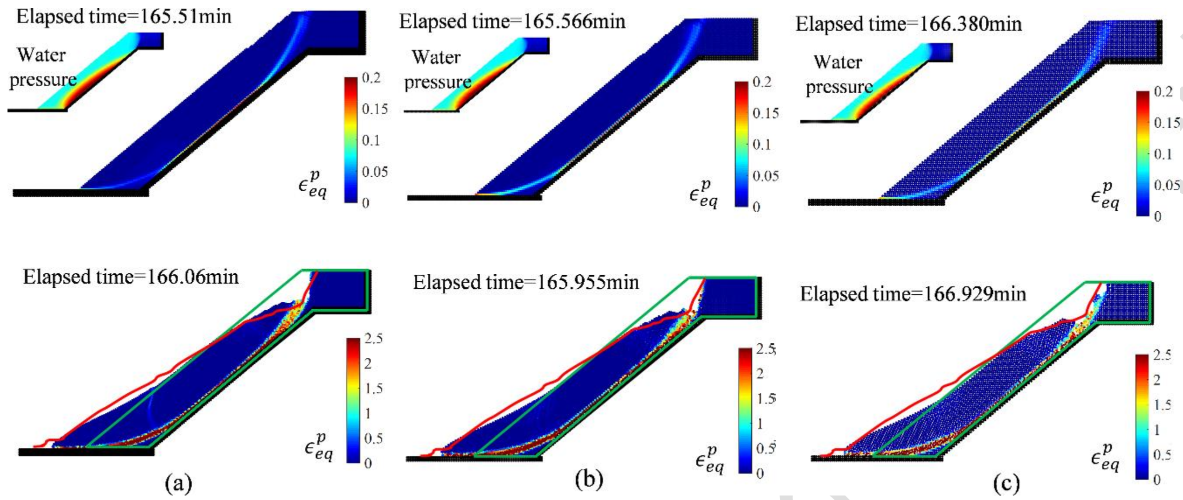

Fig. 16. SPH modelling of the rainfall-induced slope failure using different spatial discretisations: (a) dx = 4cm; (b) dx = 5cm; (c) dx = 6cm

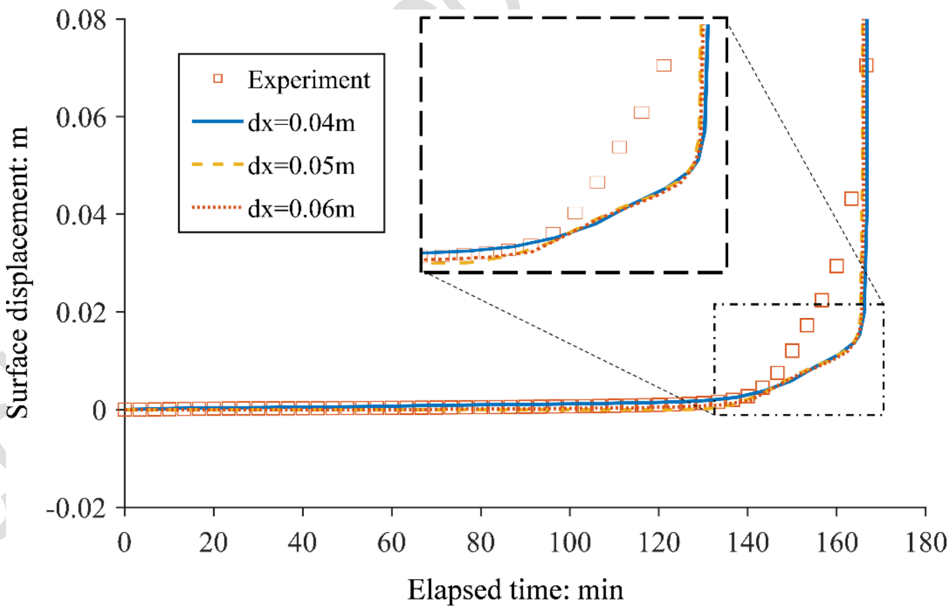

Fig. 17. Influence of spatial discretisations on the SPH prediction of ground surface displacement against the elapsed time

Influence of spatial discretisations

The mesh-dependent solution is a well-known issue of continuum-based numerical methods (e.g. FEM, MPM) when incorporating a classical strain-softening constitutive model (Bui & Nguyen, 2021). To overcome this issue, non-local plasticity theory (Bažant, 1991), gradient plasticity theory (Vardoulakis & Aifantis, 1991) or double-scale constitutive framework (Nguyen & Bui, 2020; Nguyen *et al.*, 2016) is usually adopted to provide a means to introduce a length scale into the computational model, helping to correctly scale the material responses with the size of the spatial discretisation. SPH is a continuum-based numerical method and thus, in principle, also suffers from the pathological spatial discretisation dependences. However, as pointed out by Bui & Nguyen (2021), the nature of SPH kernel approximations automatically introduces a length scale (i.e. the size of kernel function) into the SPH computational model. This "length scale" works in a similar way to that of the non-local constitutive model, thus helping SPH to mitigate its mesh-dependent issue when combined with a softening constitutive model, though it will not completely remove this issue. To demonstrate this feature of SPH, as well as the capability of the proposed SPH framework in dealing with coupled flow-large deformation problems, a series of sensitivity tests are conducted to investigate the influence of spatial discretisations on the predicted results. The above rainfall-induced slope failure test is repeated with two different particle spacing distances of $dx = 5$ cm and $dx = 6$ cm, corresponding to a total number of 3,706 and 2,626 SPH particles, respectively. Other parameters are kept the same as the previous simulation using $dx = 4$cm. Fig. 16 shows a comparison of the numerical results for three different spatial discretisations at the sliding state and the final-failure state. It can be seen that the proposed model produces very similar results for different discretisation resolutions. For instance, the predicted water table inside the slope body due to water content accumulation is almost the same in the three simulations, as shown by the contour plots in Fig 16. The failure pattern and the predicted failure surface for different spatial resolutions are very similar to each other, though some minor discrepancies could be still noticeable among the predicted results, confirming that the nature of SPH kernel interpolation mitigates the mesh-dependent issue but does not completely remove it. More importantly, the predicted time-evolution of ground surface displacement in this rainfall-induced slope failure test over a long period of time is nearly the same for different discretisation resolutions, as shown in Fig. 17. To this end, it is suggested that the proposed SPH model for rainfall-induced slope failure is less sensitive to the spatial discretisation resolutions, at least for the current studying case. To completely remove the influence of discretisation solutions on the predicted results, readers are referred to alternative approaches discussed in Bui & Nguyen (2021), which are beyond the scope of this work.

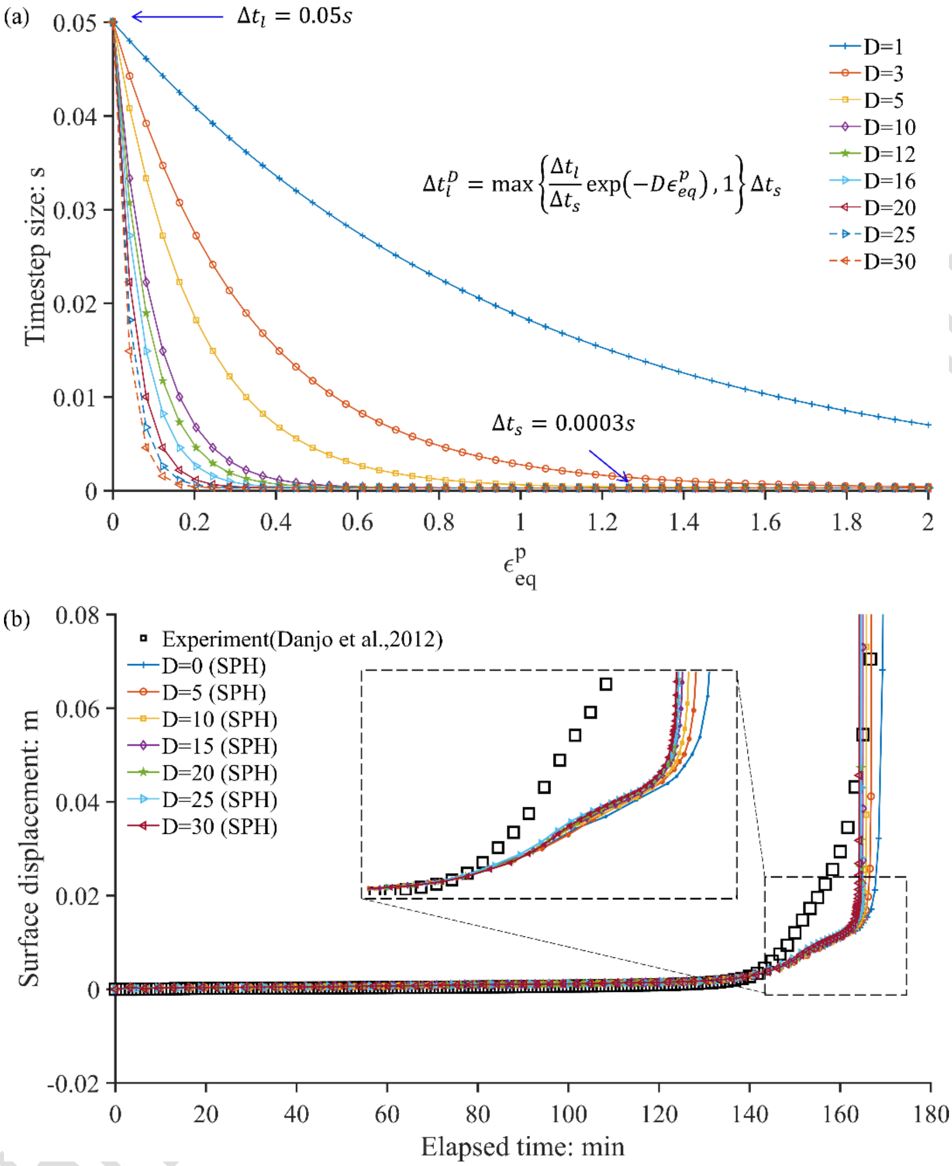

Fig. 18. Convergence analysis of the adaptive two-timestep coupling scheme: (a) Influence of parameter D on the adaptive hydraulic timestep size; (b) Influence of parameter D on the SPH prediction of surface displacement in the rainfall-induced slope failure test.

Influence of the adaptive time-stepping scheme

As discussed in the early section, an adaptive time-stepping integration scheme was proposed to facilitate the computational modelling of coupled flow-deformation problems involving the long-term diffusive process and short-term failure behaviour of granular materials. In this section, a series of sensitivity tests are conducted to investigate the influence of this technique on the predicted results. Fig. 18a shows the influence of parameter D on the evolution of the adaptive hydraulic timestep Δt_l^D and how fast this hydraulic timestep approaches the mechanical one as the deviatoric plastic shear increases. Here, the adaptive hydraulic timestep Δt_l^D is shown to drop rapidly and quickly approach the mechanical timestep Δt_s as D increases. In principle, the computational result will be affected by the selection of D and will converge as D is large enough. Our aim is to minimise such influence by adopting the largest possible value of D, while still achieving reasonable accuracy. In order to investigate the validity of the proposed technique, a convergence test is conducted by repeating the above rainfall-induced slope failure simulation with a range of D values, and for the sake of convenience, this case is only conducted using 2,626 particles.

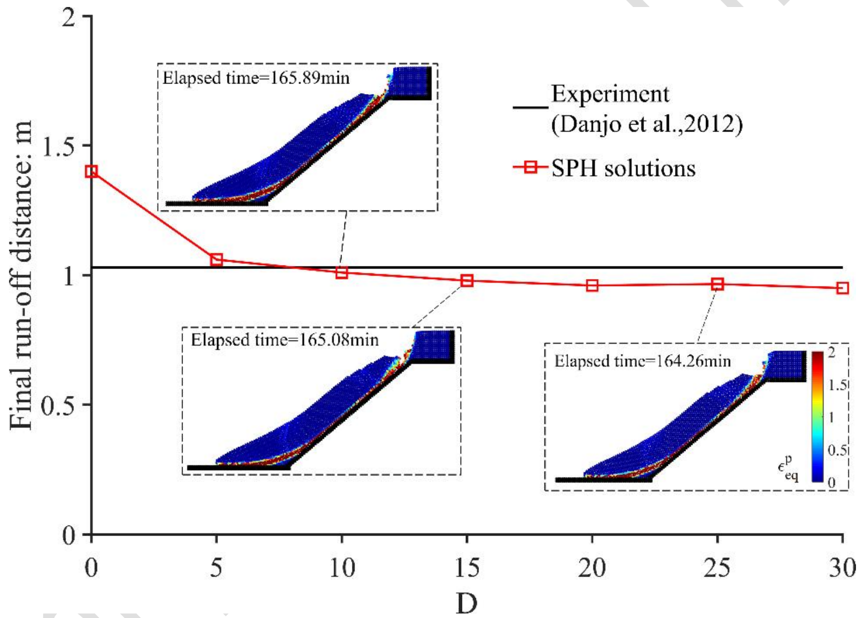

Fig. 19. Influence of the adaptive two-timestep coupling scheme on the SPH prediction of the final run-off distance of the rainfall-induced slope failure test.

Fig. 18b shows the influence of D on the predicted time-dependent surface displacement caused by the rainfall infiltration process. A similar trend in the evolution of the ground displacement is achieved for D ranging from 0 to 30, and numerical results converge as D increases beyond 25 (i.e. in a range of $\pm 1.5\%$ compared to the experiment). Some noticeable differences in the timing of the sliding or failure state, where the sliding mass detaches from the slope body and accelerates, are still observed for a smaller value of D, resulting in the delay in the timing of failure. This result is expected since a smaller value of D would delay the hydraulic timestep to approach the mechanical one, causing a lack of required coupling effect during the rapid flowing deformation of the materials. Nevertheless, if one is concerned with the precursor state or the warning state, which is very important in establishing the early warning system, it is interesting to see that the proposed adaptive time-stepping produces negligible effects. Finally, despite the above minor differences, the predicted final run-out distance shows similar results for D larger than 15 and is very close to the experimental data (see Fig. 19). More importantly, the entire simulation can be completed within less than 3 hours, instead of several days without adopting the adaptive time-stepping scheme. Overall, the above sensitive analysis suggests that the proposed adaptive time-stepping technique is reliable and produces acceptable accuracy for analyses of coupled flow-deformation problems while saving substantial computational costs.

CONCLUSION

This paper addresses one of the challenging geotechnical problems by developing a robust and advanced SPH computational framework for solving coupled flow-deformation problems in partially saturated porous media undergoing large deformations and failures. Compared to existing coupled SPH approaches, the newly proposed SPH framework is unique in the sense that it solves governing equations of multi-phase mixtures using a single set of Lagrangian particles, which is more computationally efficient and cheaper for large-scale applications. The paper demonstrates that, with a standard SWCC and a simple suction-dependent elastoplastic softening constitutive model, the proposed SPH framework could capture well the coupled behaviour of rainfall-induced transient seepage flow and subsequent slope failure process observed in the experiment. The framework is generic, given it can be used with any existing constitutive models for unsaturated soils. Furthermore, a robust adaptive time-stepping integration scheme was also proposed to address the existing computational challenge in modelling coupled flow-deformation problems involving a long-term seepage diffusion process and short-term failure behaviour of soils. It is demonstrated that the proposed adaptive time-stepping technique facilitates SPH computations for solving coupled-flow deformation problems by saving huge computational costs required to describe the long-term seepage diffusion process while still proving reasonable accuracy when it comes to the predictions of rapid flow deformation. Further advancements to the proposed SPH framework could be made by explicitly solving the evolution of airflow in the unsaturated porous media, and this will be addressed in future publications.

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APPENDIX A

The general stress-strain relationship of the suction-dependent elastic-plastic DP softening model can be derived by considering the Taylor-series expansion of the yield surface at the correct stress state around the trial stress state:

$$f(\boldsymbol{\sigma}', \kappa_c) = f^{trial} - \frac{\partial f}{\partial \boldsymbol{\sigma}'} : d\boldsymbol{\sigma}^{p'} + \frac{\partial f}{\partial \epsilon_{eq}^{p}} d\varepsilon_{eq}^{p} + \frac{\partial f}{\partial p_s} dp_s = 0 \tag{A. 1}$$

where the incremental form of the accumulated equivalent strain is given as:

$$d\varepsilon_{eq}^{p} = d\lambda \sqrt{\frac{2}{3} \frac{\partial g}{\partial \boldsymbol{s}} : \frac{\partial g}{\partial \boldsymbol{s}}} \tag{A. 2}$$

where $\boldsymbol{s}$ is the deviatoric shear stress tensor. Expanding Eq. (A. 1), which leads to:

$$f^{trial} = \frac{\partial f}{\partial \sigma'} : (\boldsymbol{D}^e : d\boldsymbol{\epsilon}^p) - \frac{\partial f}{\partial \varepsilon_{eq}^{p}} d\varepsilon_{eq}^{p} - \frac{\partial f}{\partial p_s} : dp_s$$

$$\Rightarrow f^{trial} = \frac{\partial f}{\partial \boldsymbol{\sigma}'} : \left(\boldsymbol{D}^e : d\lambda \frac{\partial g}{\partial \boldsymbol{\sigma}'} \right) - \frac{\partial f_{N+1}^{trial}}{\partial \varepsilon_{eq}^{p}} d\lambda \sqrt{\frac{2}{3} \frac{\partial g}{\partial \boldsymbol{s}} : \frac{\partial g}{\partial \boldsymbol{s}}} - \frac{\partial f}{\partial p_s} : dp_s \tag{A. 3}$$

Re-arrange Eq. (A. 3), leading to the expression of the plastic multiplier $d\lambda$:

$$d\lambda = \frac{f^{trial} + \dfrac{\partial f}{\partial p_s} dp_s}{H_e + H_p} \tag{A. 4}$$

where

$$\frac{\partial f}{\partial \boldsymbol{\sigma}'} = \xi_\phi \boldsymbol{\delta} + \frac{\boldsymbol{s}}{2\sqrt{J_2}} \quad and \quad \frac{\partial g}{\partial \boldsymbol{\sigma}'} = \xi_\psi \boldsymbol{\delta} + \frac{\boldsymbol{s}}{2\sqrt{J_2}} \quad and \quad \boldsymbol{s} = \boldsymbol{\sigma}' - \frac{I_1}{3}\boldsymbol{\delta}$$

$$\frac{\partial f}{\partial p_s} = \frac{\partial f}{\partial \phi}\frac{\partial \phi}{\partial \phi_s}\frac{\partial \phi_s}{\partial p_s} + \frac{\partial f}{\partial c}\frac{\partial c}{\partial c_s}\frac{\partial c_s}{\partial p_s}$$

$$H_p = -\frac{\partial f}{\partial \varepsilon_{eq}^p}\sqrt{\frac{2}{3}\frac{\partial g}{\partial \boldsymbol{s}}:\frac{\partial g}{\partial \boldsymbol{s}}}$$

$$He = \frac{\partial f}{\partial \boldsymbol{\sigma}'}:\mathbf{D_e}:\frac{\partial g}{\partial \boldsymbol{\sigma}'}$$

$$\frac{\partial f}{\partial \epsilon_{eq}^p} = \frac{\partial f}{\partial c}\frac{\partial c}{\partial \epsilon_{eq}^p} + \frac{\partial f}{\partial \phi}\frac{\partial \phi}{\partial \epsilon_{eq}^p}$$

$$\frac{\partial f}{\partial c} = \frac{\partial f}{\partial k_c}\frac{\partial k_c}{\partial c} = (-1)\frac{3}{\sqrt{9 + 12tan\phi^2}}$$

$$\frac{\partial f}{\partial \phi} = \frac{\partial f}{\partial k_c}\frac{\partial k_c}{\partial \phi} + \frac{\partial f}{\partial \xi_\phi}\frac{\partial \xi_\phi}{\partial \phi} =$$

$$\frac{36ctan\phi(1+\tan^2\phi)}{\left(9+12(tan\phi^2)\right)^{\frac{3}{2}}} + I_1\left(\frac{(1+\tan^2\phi)}{\sqrt{9+12tan\phi^2}} - \frac{12\tan^2\phi\,(1+\tan^2\phi)}{\left(9+12(tan\phi^2)\right)^{\frac{3}{2}}}\right)$$

$$\frac{\partial c}{\partial \epsilon_p^{eq}} = -\eta\left(c_p - c_r\right)e^{-\eta\varepsilon_{eq}^p} \quad and \quad \frac{\partial \phi}{\partial \epsilon_p^{eq}} = -\eta\left(\phi_p - \phi_r\right)e^{-\eta\varepsilon_{eq}^p}$$

$$\frac{\partial \phi_s}{\partial \phi} = \frac{A'}{p_{atm}} \quad and \quad \frac{\partial c_s}{\partial c} = c_{max}\frac{B'}{p_{atm}}e^{-B'\frac{p_s}{p_{atm}}}$$

APPENDIX B

Within the context of Terzaghi's one-dimensional consolidation theory, the analytical solution of the excess pore water pressure with respect to time is given as follows (Terzaghi, 1996):

$$p_l = \sum_{k=1}^{k=\infty}\frac{2p_{l0}}{M}\sin\left(\frac{Mz}{L}\right)\exp(-M^2T_v) \qquad \text{(B.1)}$$

where p_{l0} is the initial excess pore pressure; L is the consolidation length; $M = 0.5(2k-1)\pi$ is a normalised factor; and the dimensional time factor is given as:

$$T_v = \frac{c_v t}{L^2} \qquad \text{(B. 2)}$$

where:

$$c_v = \frac{k_{sat}}{\gamma m_v} = \frac{k^{sat}(1-v)E_s}{(1+v)(1-2v)\gamma_l}$$

where c_v is the consolidation coefficient; v is Poison's ratio; E_s is Young's modulus; and m_v is the compression index defined by:

$$m_v = \frac{(1+v)(1-2v)}{(1-v)E_s} \qquad \text{(B.3)}$$

The surface settlement of the soil column can be obtained using the average degree of consolidation, which is given as follows:

$$S_{T_v} = U_a S_f \qquad \text{(B. 4)}$$

where

$$U_a = 1 - \sum_{k}^{k=\infty} \frac{2}{M^2} \exp(-M^2 T_v)$$

where S_f is the final surface settlement in the column after the excess pore water pressure is fully dissipated and could be obtained as follows:

$$S_f = q_0 m_v L \qquad \text{(B. 5)}$$

where q_0 is the surcharge load; and m_v is the compression index.

NOTATION

A'	Parameter controlling the variation of friction angle
$\hat{\alpha}$	The compressibility of soil skeleton (kPa^{-1})
α_d	The dimensional normalising factor
$\hat{\beta}$	The compressibility of the water phase (kP_a^{-1})
B'	Parameter controlling the variation of cohesion
b	The gravitational force vector
c_{max}	The maximum cohesion suction bring-up (*kPa*)
c, c', c_s	Total cohesion, saturated cohesion, suction dependent cohesion (*kPa*)
c_{sp}	Sound speed in the material (*m/s*)
c_p, c_r	The peak and residual cohesion (*kPa*)
c_v	The consolidation coefficient
C_l	The specific storage term in unsaturated soil (m^{-1})

C_s	The specific moisture term (m^{-1})
C_k	The Kozeny-Carman coefficient
$\mathbf{C}_{ij}$	A stabilisation term in SPH
$\mathbf{D}^e$	The elastic stiffness tensor (kPa)
δ^{mn}	Kronecker delta
$\tilde{F}_{ij}$	The scalar part of the normalised kernel gradient
E_s	Young's modulus of solid material (Pa)
e	The void ratio
ε_p^{eq}	The accumulated equivalent plastic strain
$\boldsymbol{\varepsilon}^p$	The plastic strain tensor
F	Timestep adapting coefficient
f	The yield surface
f^{trial}	The yield surface with a trial stress
g	The plastic potential
ξ	A coefficient for global dumping technique
ξ_ϕ, ξ_ψ	The DP constant
ϕ, ϕ', ϕ_s	Total friction angle, saturated friction angle, suction dependent friction angle (°)
ϕ_p, ϕ_r	Peak friction angle, residual friction angle (°)
ψ, ψ_0	The dilation angle , the initial dilation angle (°)
η	Softening controlling coefficient
ρ_a, ρ_l, ρ_s	Intrinsic mass density of air phase, water phase, solid phase (kg/m^3)
p_{atm}	The atmospheric pressure (kPa)
p_a, p_l, p_s	Pore-air pressure, pore-water pressure, suction (kg/m^3)
$\bar{\rho}_a$	Partial density of air phase (kg/m^3)
$\bar{\rho}_l$	Partial density of water phase (kg/m^3)
$\bar{\rho}_s$	Partial density of solid phase (kg/m^3)
q	The normalised distance (m)
Q	The mixed volumetric stiffness of the mixture, (kPa)
ρ_t	Total density of the mixture (kg/m^3)
g_a	Material parameter of SWRC (m^{-1})
g_n	Material parameter of SWRC

g_c	Empirical parameters of SWRC
H_p	The softening modulus
H	Total water head (m)
h	Water pressure head (m)
h_{sml}	The smoothing length (m)
m_v	The coefficient of volume change (kPa^{-1})
$\mathbf{I}$	The identity matrix
K_S	The bulk modulus of the soil skeleton (kPa)
K_l^{sat}	The bulk modulus of water phase (kPa)
$\mathbf{k}_a$	The permeability tensor of air phase (m/s)
κ_c	DP constant
$\mathbf{k}_l$	The permeability tensor of water phase (m/s)
k_{sat}	The saturated permeability of water phase (m/s)
$\mathbf{L}_{ij}$	The normalised matrix
$d\lambda$	The plastic multiplier
m^t	Total mass (kg)
m_v	Compression index, (1/kPa)
n_a, n_l, n_s	Void fraction of air phase, water phase, solid phase
n	Porosity of the mixture
$\mathbf{r}_{ji}$	Distance between particle i and j (m)
$\mathbf{R}^{\alpha\beta}$	The drag force vector between phase α and β (N)
S_r	The degree of saturation
S_{res}	The residual degree of saturation
S_{sat}	The saturated degree of saturation
S_f	Final surface settlement
S_{T_v}	Surface settlement at T_v
$\boldsymbol{\sigma}$	Total stress tensor of the mixture (Pa)
σ_v	Vertical total stress (Pa)
$\boldsymbol{\sigma}'$	Effective stress tensor of the solid phase (Pa)
$\boldsymbol{\sigma}_a, \boldsymbol{\sigma}_l, \boldsymbol{\sigma}_s$	Intrinsic stress tensor of air, water, solid phases (Pa)
$\bar{\boldsymbol{\sigma}}_a, \bar{\boldsymbol{\sigma}}_l, \bar{\boldsymbol{\sigma}}_s$	Partial stress tensors of air, water and solid phases (Pa)
T_v	The dimensionless time factor

$\Delta t_s, \Delta t_l$	The size of mechanics timestep and fluid flow timestep (s)
Δt_l^D	The adapted timestep size for fluid flow (s)
$\boldsymbol{x}$	Position vector (m)
$\frac{d^{\alpha}(\quad)}{dt}$	The material derivative on component α (s^{-1})
v	Poisson's ratio
V_j	The volume occupied by particle j (m^2)
$\mathbf{v}_s$	Velocity tensor of the solid phase ($\mathrm{m/s}$)
W	The symmetric smoothing kernel function
$\overline{\mathbf{w}}_{ls}$	The relative velocity between water and solid phase ($\mathrm{m/s}$)
$\overline{\mathbf{w}}_{as}$	The relative velocity between air and solid phase ($\mathrm{m/s}$)
z	Elevation (m)
$\langle \quad \rangle$	The kernel approximation operator

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